DESIGN DATA BOOK — SCREWED JOINTS & FASTENERS

Standard Engineering Design Data, Formulas & Solved Protocols (IS: 4218 - 1976)

Fastener Joint Classification & Design Index

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SECTION REF 1: Table 11.1 — Standard Screw Threads (Coarse Series)

Designation	Pitch p	Nominal d	Pitch d_p	Core d_c	Depth h	Stress Area A_c
M 4	0.7 mm	4.0 mm	3.545 mm	3.141 mm	0.429 mm	8.78 mm²
M 5	0.8 mm	5.0 mm	4.480 mm	4.019 mm	0.491 mm	14.2 mm²
M 6	1.0 mm	6.0 mm	5.350 mm	4.773 mm	0.613 mm	20.1 mm²
M 8	1.25 mm	8.0 mm	7.188 mm	6.466 mm	0.767 mm	36.6 mm²
M 10	1.5 mm	10.0 mm	9.026 mm	8.160 mm	0.920 mm	58.3 mm²
M 12	1.75 mm	12.0 mm	10.863 mm	9.858 mm	1.074 mm	84.0 mm²
M 14	2.0 mm	14.0 mm	12.701 mm	11.546 mm	1.227 mm	115 mm²
M 16	2.0 mm	16.0 mm	14.701 mm	13.546 mm	1.227 mm	157 mm²
M 18	2.5 mm	18.0 mm	16.376 mm	14.933 mm	1.534 mm	192 mm²
M 20	2.5 mm	20.0 mm	18.376 mm	16.933 mm	1.534 mm	245 mm²
M 22	2.5 mm	22.0 mm	20.376 mm	18.933 mm	1.534 mm	303 mm²
M 24	3.0 mm	24.0 mm	22.051 mm	20.320 mm	1.840 mm	353 mm²
M 27	3.0 mm	27.0 mm	25.051 mm	23.320 mm	1.840 mm	459 mm²
M 30	3.5 mm	30.0 mm	27.727 mm	25.706 mm	2.147 mm	561 mm²
M 33	3.5 mm	33.0 mm	30.727 mm	28.706 mm	2.147 mm	694 mm²
M 36	4.0 mm	36.0 mm	33.402 mm	31.093 mm	2.454 mm	817 mm²
M 42	4.5 mm	42.0 mm	39.077 mm	36.416 mm	2.760 mm	1104 mm²
M 48	5.0 mm	48.0 mm	44.752 mm	41.795 mm	3.067 mm	1465 mm²
M 52	5.0 mm	52.0 mm	48.752 mm	45.795 mm	3.067 mm	1755 mm²
M 56	5.5 mm	56.0 mm	52.428 mm	49.177 mm	3.067 mm	2022 mm²

SECTION REF 2: Table 11.1 (Fine Series) & Table 11.2 (Joint Constants K)

1. Table 11.1 — Fine Series Screw Threads

Designation	Pitch p	Nominal d	Pitch d_p	Core d_c	Stress Area A_c
M 8 x 1	1.0 mm	8.0 mm	7.350 mm	6.773 mm	39.2 mm²
M 10 x 1.25	1.25 mm	10.0 mm	9.188 mm	8.466 mm	61.6 mm²
M 12 x 1.25	1.25 mm	12.0 mm	11.184 mm	10.466 mm	92.1 mm²
M 14 x 1.5	1.5 mm	14.0 mm	12.160 mm	125 mm²	125 mm²
M 16 x 1.5	1.5 mm	16.0 mm	15.026 mm	14.160 mm	167 mm²
M 20 x 1.5	1.5 mm	20.0 mm	19.026 mm	18.160 mm	272 mm²
M 24 x 2	2.0 mm	24.0 mm	22.701 mm	21.546 mm	384 mm²

2. Table 11.2 — Joint Constants & Gasket Factors (K)

Type of Joint / Gasket Connection	Gasket Factor Value (K)
Metal to metal joint with through bolts	0.00 to 0.10
Hard copper gasket with long through bolts	0.25 to 0.50
Soft copper gasket with long through bolts	0.50 to 0.75
Soft packing with through bolts	0.75 to 1.00
Soft packing with studs	1.00

Empirical Core Diameter Rule (When Tables are Unavailable):

For standard metric coarse series fasteners:

Core Diameter: $d_c \approx 0.84d$

Core Stress Area: $A_c \approx \frac{\pi}{4}(0.84d)^2$

SECTION 1: Standard Tensile Capacity Analysis (M 30 Fasteners)

System Parameters

Nominal Bolt Diameter: $d = 30\ \mathrm{mm}$ (M 30 Coarse Series)

Safe Tensile Stress: $\sigma_t = 42\ \mathrm{MPa} = 42\mathrm{N}/\mathrm{mm}^2$

Preload $F_i = 0$

Design Protocol

Retrieve core stress area A_c from Table 11.1

Calculate safe tensile load capacity $P = A_c \sigma_t$

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1. Core Stress Area Lookup (A_c)

Table 11.1 lookup for M 30 coarse series thread

$$A_c = f(d)$$
$$= 561\mathrm{mm}^2$$

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2. Safe Tensile Load Capacity (P)

Maximum allowable axial tension force without initial tightening

$$P = A_c \times \sigma_t$$
$$= 561\mathrm{mm}^2 \times 42\mathrm{N}/\mathrm{mm}^2$$
$$= \mathbf{23.562\ kN}$$

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3. Design Output

Safe Tensile Load $P = 23.562\ \mathrm{kN}$

SECTION 2: Initial Tightening Stress & Preload Analysis (Tap Bolts)

System Parameters

Nominal Tap Bolt Size: $d = 24$ mm (M 24 Coarse Series)

Clamped Rigid Machine Joint

Neglect external separation load

Design Protocol

Lookup core root diameter d_c from Table 11.1

Calculate initial tightening load $P = 2840d$

Calculate induced tensile stress $\sigma_t = \frac{P}{\frac{\pi}{4}d_c^2}$

---

1. Core Root Diameter Lookup (d_c)

Table 11.1 (coarse series) lookup for M 24 thread

$$d_c = 20.32 \text{ mm}$$

---

2. Initial Tightening Load (P)

Empirical initial tightening tension equation

$$P = 2840 \cdot d$$
$$= 2840 \times 24$$
$$= 68160 \text{ N}$$

3. Induced Tightening Stress (σ_t)

Initial tension force balance equation

$$68160 = \frac{\pi}{4}(20.32)^2 \cdot \sigma_t$$

$$\sigma_t = \frac{68160}{324}$$

$$= \mathbf{210\ MPa}$$

4. Design Output

Tightening Stress $\boldsymbol{\sigma_t = 210\ \mathrm{MPa}}$

SECTION 3: Lifting Eye Bolt Sizing & Thread Selection

System Parameters

Lifting Load: $P = 60 \text{ kN} = 60000 \text{ N}$

Permissible Tensile Stress: $\sigma_t = 100 \text{ MPa} = 100 \text{N/mm}^2$

Thread Series: ISO Metric Coarse

Design Protocol

Equate lifting load to tensile capacity $P = \frac{\pi}{4} d_c^2 \sigma_t$

Solve required core root diameter d_c

Select standard bolt from Table 11.1

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1. Required Core Root Diameter (d_c)

Axial tension load equation for lifting eye bolt

$$P = \frac{\pi}{4} (d_c)^2 \cdot \sigma_t$$

$$60000 = \frac{\pi}{4} (d_c)^2 \times 100$$

$$d_c = 27.64 \text{ mm}$$

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2. Standard Thread Selection (Table 11.1)

Select standard coarse series thread

$$\begin{aligned} d_{\text{c, std}} &\geq d_c \\ &= 27.64 \text{ mm} \\ d_c &= 28.706 \text{ mm} \\ d &= 33 \text{ mm} \end{aligned}$$

3. Design Output	**Select Fastener Designation: M 33 Bolt ($d = 33$ mm, $d_c = 28.706$ mm)**

SECTION 3: Lifting Eye Bolt — MECHANICAL DIAGRAM

11.13 Stress due to Combined Forces

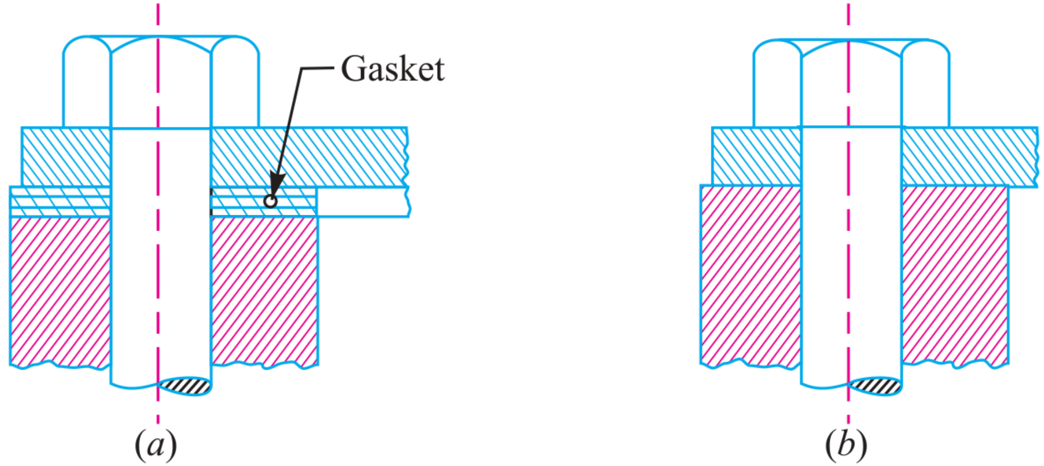

Fig. 11.23

Figure 11.23: Structural Lifting Eye Bolt Configuration and Load Path

SECTION 4: Flanged Shaft Coupling Fastener Sizing under Torsional Shear

System Parameters

Transmitted Torque: $T = 25\ \mathrm{N \cdot m} = 25000\ \mathrm{N \cdot mm}$

Pitch Circle Radius: $R_p = 30\ \mathrm{mm}$

Bolt Count: $n = 4$ | Allowable Shear Stress: $\tau = 30\ \mathrm{MPa} = 30 \mathrm{N/mm}^2$

Design Protocol

Calculate total tangential shearing load $P_s = \frac{T}{R_p}$

Equate P_s to shear capacity of n bolts $P_s = n(\frac{\pi}{4} d_c^2)\tau$

Select standard bolt size from Table 11.1

1. Total Shearing Load (P_s)

Total tangential shearing force acting at PCD

$$P_s = \frac{T}{R_p}$$

$$= \frac{25000}{30}$$

$$= 833.3\ \mathrm{N}$$

2. Required Core Diameter (d_c)

Solve root diameter for $n = 4$ bolts carrying shear

$$P_s = n \cdot \left[\frac{\pi}{4}(d_c)^2\right] \cdot \tau$$

$$833.3 = 4 \cdot \left[\frac{\pi}{4}(d_c)^2\right] \times 30$$

$$d_c = 2.97 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse series thread

$$d_{\text{c, std}} \geq d_c$$

$$= 2.97 \text{ mm}$$

$$d_c = 3.141 \text{ mm}$$

$$d = 4 \text{ mm}$$

4. Design Output

Select Fastener Designation: M 4 Bolt ($d = 4 \text{ mm}, d_c = 3.141 \text{ mm}$)

SECTION 5: Safety Valve Fulcrum Pin Sizing & Lever Equilibrium

System Parameters

Valve Diameter: $D = 100\ \mathrm{mm}$

Blow-off Steam Pressure: $p = 1.6 \mathrm{N/mm}^2$

Leverage Ratio: $\frac{L_1}{L_2} = 8$ | Permissible Tensile Stress: $\sigma_t = 50\ \mathrm{MPa}$

Design Protocol

Calculate upward steam force $F = \frac{\pi}{4} D^2 p$

Apply lever moment equilibrium for end load W and fulcrum load $P = F - W$

Size fine metric thread d_c for fulcrum pin

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1. Total Upward Force on Valve (F)

Total steam pressure load on valve disc

$$F = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4} (100)^2 \times 1.6$$

$$= 12568\ \mathrm{N}$$

2. Lever End Load (W) & Fulcrum Load (P)

Static moment equilibrium for Type 1 lever

$$W = \frac{F}{\frac{L_1}{L_2}}$$

$$= \frac{12568}{8}$$

$$= 1571 \text{ N}$$

$$P = F - W$$

$$= 12568 - 1571$$

$$= 10997 \text{ N}$$

3. Required Core Diameter (d_c)

Solve root core diameter for fulcrum screw

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$10997 = \frac{\pi}{4}(d_c)^2 \times 50$$

$$d_c = 16.7 \text{ mm}$$

4. Standard Fine Thread Selection (Table 11.1)

Select fine series metric thread

$$d_{\text{c, std}} \geq d_c$$

$$= 16.7 \text{ mm}$$

$$d_c = 18.376 \text{ mm}$$

$$\text{Fine thread: M } 20 \times 1.5$$

5. Design Output	**Select Fine Series Thread: M 20 x 1.5 ($d = 20\ \mathrm{mm}, d_c = 18.376\ \mathrm{mm}$)**

SECTION 6: Cylinder Head Cover Stud Layout & Leak-Proof Pitch Verification

System Parameters

Cylinder Effective Diameter: $D = 350 \text{ mm}$

Maximum Steam Pressure: $p = 1.25 \text{N/mm}^2$

Permissible Stud Stress: $\sigma_t = 33 \text{ MPa} = 33 \text{N/mm}^2$

Design Protocol

Calculate total upward steam load $P = \frac{\pi}{4} D^2 p$

Assume M 24 studs ($d_c = 20.32 \text{ mm}$) to find count $n = \frac{P}{P_1}$

Verify leak-proof pitch limits $20\sqrt{d_1} \leq p_c \leq 30\sqrt{d_1}$

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1. Total Upward Steam Load (P)

Total gas pressure load on cylinder cover

$$P = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4} (350)^2 \times 1.25$$

$$= 120265 \text{ N}$$

2. Capacity per M 24 Stud (P_1) & Stud Count (n)

Resisting force capacity per M 24 stud

$$P_1 = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$= \frac{\pi}{4}(20.32)^2 \times 33$$

$$= 10700 \text{ N}$$

$$n = \frac{P}{P_1}$$

$$= \frac{120265}{10700}$$

$$= 11.24$$

$$\Rightarrow \boldsymbol{n = 12} \textbf{ studs}$$

3. Pitch Circle Diameter (D_p) & Circumferential Pitch (p_c)

Pitch circle geometry

$$D_p = D + 2t + 3d_1$$

$$= 350 + 2(10) + 3(25)$$

$$= 445 \text{ mm}$$

$$p_c = \frac{\pi D_p}{n}$$

$$= \frac{\pi \times 445}{12}$$

$$= 116.5 \text{ mm}$$

4. Leak-Proof Pitch Verification

Empirical leak-tight pitch limits

$$p_c, \text{min} = 20\sqrt{d_1}$$
$$= 20\sqrt{25}$$
$$= 100 \text{ mm}$$

$$p_c, \text{max} = 30\sqrt{d_1}$$
$$= 30\sqrt{25}$$
$$= 150 \text{ mm}$$

$$100 \text{ mm} \leq 116.5 \text{ mm} \leq 150 \text{ mm} \ (\textbf{SAFE} \ \& \ \textbf{SATISFACTORY})$$

5. Design Output

Use 12 Studs of M 24 Size ($n = 12$, M 24 Studs)

SECTION 6: Cylinder Cover Stud Layout — MECHANICAL DIAGRAM

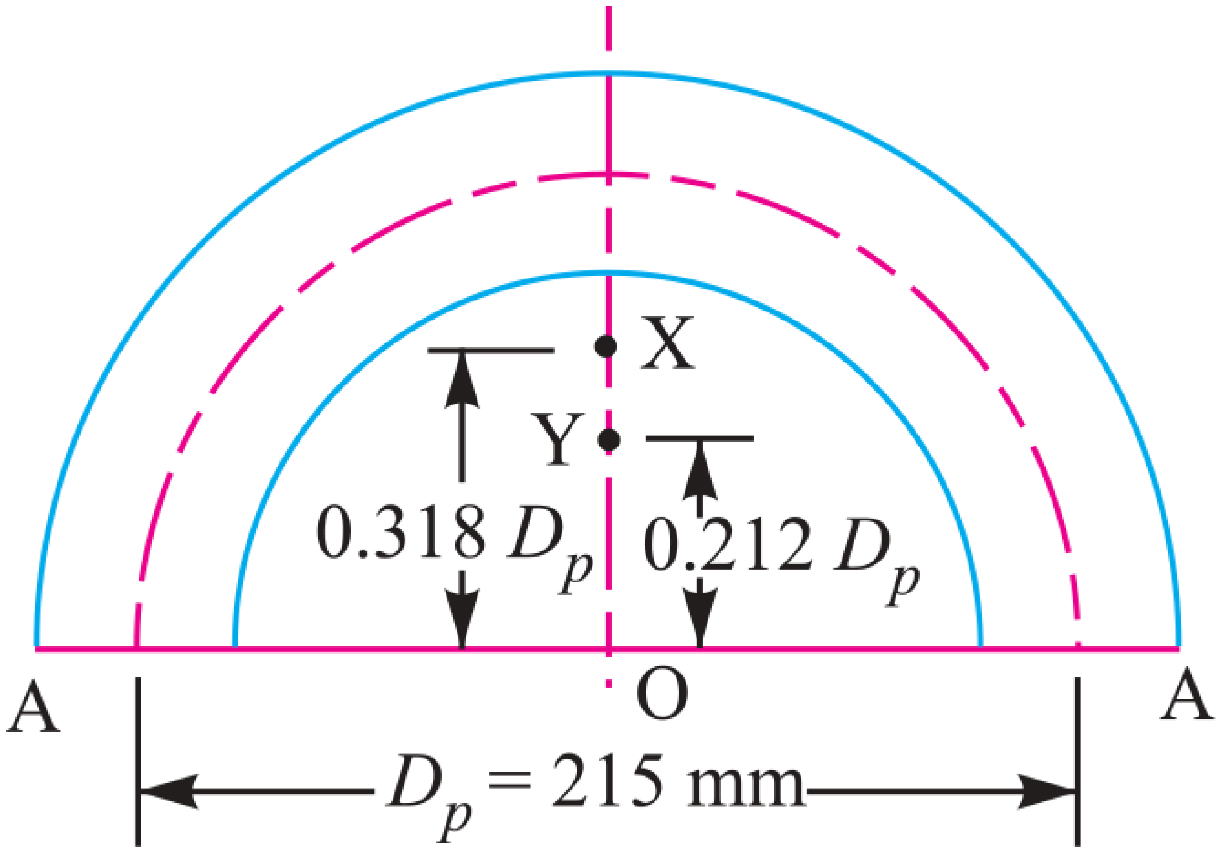

Fig. 11.27

Figure 11.27: Cylinder Head Joint & Stud Pitch Circle Assembly

SECTION 7: Pressure Vessel Inspection Cover Plate & Bolt Sizing

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System Parameters

Inspection Hole Diameter: $D = 120$ mm ($r = 60$ mm)

Internal Vessel Pressure: $p = 6\mathrm{N/mm}^2$

Plate Tensile Stress: $\sigma_t = 60$ MPa | Bolt Stress: $\sigma_{\mathrm{tb}} = 40$ MPa

Design Protocol

Part 1: Calculate shell wall thickness t via Lame

Part 2: Determine bolt count n for M 24 and check pitch p_c

Part 3: Calculate cover plate thickness t_1 under bending moment

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1. Shell Wall Thickness (t)

Lame equation for thick pressure vessels

$$t = r\left[\sqrt{\frac{\sigma_t + p}{\sigma_t - p}} - 1\right]$$

$$= 60\left[\sqrt{\frac{60+6}{60-6}} - 1\right]$$

$$= 6.3 \text{ mm}$$

Adopt Shell Thickness $t = 10$ mm

2. Total Upward Force (P)

Upward pressure force acting on inspection cover plate

$$P = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4} (120)^2 \times 6$$

$$= 67860 \text{ N}$$

3. Resisting Force per Bolt (P_1)

Capacity per M 24 bolt ($d_c = 20.32$ mm)

$$P_1 = \frac{\pi}{4} (d_c)^2 \cdot \sigma_{\text{tb}}$$

$$= \frac{\pi}{4} (20.32)^2 \times 40$$

$$= 12973 \text{ N}$$

4. Required Bolt Count (n)

Number of M 24 fixing bolts required

$$n = \frac{P}{P_1}$$

$$= \frac{67860}{12973}$$

$$= 5.23$$

$$\Rightarrow \boldsymbol{n = 6} \textbf{ bolts}$$

5. Pitch Circle Diameter (D_p)

Hole diameter $d_1 = 25$ mm for M 24 bolts

$$D_p = D + 2t + 3d_1$$

$$= 120 + 2(10) + 3(25)$$

$$= 215 \text{ mm}$$

6. Circumferential Pitch (p_c)

Pitch spacing for $n = 6$ bolts on PCD

$$p_c = \frac{\pi D_p}{n}$$
$$= \frac{\pi \times 215}{6}$$
$$= 112.6 \text{ mm}$$

7. Minimum Pitch Limit (p_c, min)

Empirical lower limit $20\sqrt{d_1}$

$$p_c, \text{min} = 20\sqrt{d_1}$$
$$= 20\sqrt{25}$$
$$= 100 \text{ mm}$$

8. Maximum Pitch Limit (p_c, max)

Empirical upper limit $30\sqrt{d_1}$

$$p_c, \text{max} = 30\sqrt{d_1}$$
$$= 30\sqrt{25}$$
$$= 150 \text{ mm}$$

9. Leak-Proof Pitch Verification

Pitch range verification

$100 \text{ mm} \leq 112.6 \text{ mm} \leq 150 \text{ mm}$ ($\textbf{SAFE}$ & $\textbf{SATISFACTORY}$)

10. Selected Bolt Designation

Pitch 112.6 mm is within 100 mm to 150 mm limit

Size of Bolt = M 24 Fasteners ($d = 24$ mm, $d_c = 20.32$ mm)

11. Cover Plate Bending Moment (M)

Bending moment acting at central section A-A

$$M = 0.053PD_p$$
$$= 0.053 \times 67860 \times 215$$
$$= 773265 \text{ N} \cdot \text{mm}$$

12. Outside Diameter of Cover (D_o)

Outer plate boundary diameter

$$D_o = D_p + 3d_1$$
$$= 215 + 3(25)$$
$$= 290 \text{ mm}$$

13. Plate Bending Width (w)

Net plate width carrying bending

$$w = D_o - 2d_1$$
$$= 290 - 2(25)$$
$$= 240 \text{ mm}$$

14. Cover Section Modulus (Z)

Bending section modulus of plate

$$Z = \frac{1}{6}w(t_1)^2$$
$$= \frac{1}{6}(240)(t_1)^2$$
$$= 40(t_1)^2$$

15. Required Cover Thickness (t_1)

Solved thickness of cover plate

$$\sigma_t = \frac{M}{Z}$$

$$60 = \frac{773265}{40(t_1)^2}$$

$$(t_1)^2 = \frac{773265}{40 \times 60}$$

$$= 322$$

$$t_1 = \mathbf{18\ mm}$$

16. Design Output

Final Design: 6 Bolts of M 24 Size, Cover Plate Thickness $\boldsymbol{t_1 = 18\ \mathrm{mm}}$

SECTION 8: Preloaded Flange Fasteners with Gasket Compression (Soft Copper)

System Parameters

Steam Pressure: $p = 0.7 \mathrm{N/mm}^2$

Number of Bolts: $n = 12$

Effective Cylinder Diameter: $D = 300 \text{ mm}$

Allowable Stress: $\sigma_t = 100 \text{ MPa}$ | Soft Copper Gasket Ratio: $K = 0.5$

Design Protocol

Calculate external load per bolt $P_2 = \frac{P_{\text{total}}}{12}$

Formulate resultant axial load $P = 2840d + KP_2$

Solve governing quadratic for nominal diameter d

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1. Total External Load on Cylinder Head (P_{total})

Upward steam pressure force on 12 bolts

$$P_{\text{total}} = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4}(300)^2 \times 0.7$$

$$= 49490 \text{ N}$$

2. External Load per Bolt (P_2)

External load shared per bolt

$$P_2 = \frac{P_{\text{total}}}{n}$$
$$= \frac{49490}{12}$$
$$= 4124 \text{ N}$$

3. Initial Tightening Tension (P_1)

Empirical tightening tension formula

$$P_1 = 2840d \text{ N} \quad (d \text{ in mm})$$

4. Gasket Compression Factor (K)

Table 11.2 for soft copper gasket with long through bolts

Minimum Gasket Factor $K = 0.5$

5. Resultant Axial Load per Bolt (P)

Combined preload and gasket compression load

$$P = P_1 + K \cdot P_2$$
$$= 2840d + 0.5 \times 4124$$
$$= (2840d + 2062) \text{ N}$$

6. Resisting Load Capacity per Bolt (P)

Tensile strength with core diameter $d_c = 0.84d$

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$= \frac{\pi}{4}(0.84d)^2 \times 100$$

$$= 55.4d^2$$

7. Governing Quadratic Equation for Size (d)

Equating resultant load to resisting load

$$55.4d^2 = 2840d + 2062$$

$$55.4d^2 - 2840d - 2062 = 0$$

$$d^2 - 51.3d - 37.2 = 0$$

8. Quadratic Solution for Nominal Diameter (d)

Quadratic formula evaluation

$$d = \frac{51.3 + \sqrt{(51.3)^2 + 4(37.2)}}{2}$$

$$= \frac{51.3 + 52.7}{2}$$

$$= 52 \text{ mm}$$

9. Design Output / Selected Bolt

Size of Bolt = M 52 Fasteners ($d = 52$ mm)

SECTION 9: Soderberg Fatigue & Preload Design Criterion

System Parameters

$D = 300 \text{ mm}, p = 1.5\text{N/mm}^2, n = 8, \sigma_y = 330 \text{ MPa}, \sigma_e = 240 \text{ MPa}$

Initial Preload: $P_1 = 1.5P_2$, Gasket Factor: $K = 0.5$, "F.S." = 2

Design Protocol

Calculate maximum load $P_1 + KP_2$ & minimum load P_1 per bolt

Determine mean load P_m and variable load P_v per bolt

Apply Soderberg fatigue equation to solve core diameter d_c

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1. Total Steam Load (P_2) & Initial Preload (P_1)

Steam pressure force and initial preload tension

$$P_2 = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4}(300)^2 \times 1.5$$

$$= 106040 \text{ N}$$

$$P_1 = 1.5P_2$$

$$= 1.5 \times 106040$$

$$= 159060 \text{ N}$$

2. Maximum & Minimum Load per Bolt

Load allocation shared across $n = 8$ bolts

$$P_{\text{max}} = P_1 + KP_2$$
$$= 159060 + 0.5(106040)$$
$$= 212080 \text{ N}$$
$$P_{\text{max, bolt}} = \frac{212080}{8}$$
$$= 26510 \text{ N}$$
$$P_{\text{min, bolt}} = \frac{159060}{8}$$
$$= 19882 \text{ N}$$

3. Mean (P_m) & Variable (P_v) Loads per Bolt

Mean and alternating cyclic load components

$$P_m = \frac{P_{\text{max, bolt}} + P_{\text{min, bolt}}}{2}$$
$$= \frac{26510 + 19882}{2}$$
$$= 23196 \text{ N}$$
$$P_v = \frac{P_{\text{max, bolt}} - P_{\text{min, bolt}}}{2}$$
$$= \frac{26510 - 19882}{2}$$
$$= 3314 \text{ N}$$

4. Mean (σ_m) & Variable (σ_v) Stresses

Stress area $A_c = \frac{\pi}{4}(d_c)^2 = 0.7854(d_c)^2$

$$\sigma_m = \frac{P_m}{A_c}$$

$$= \frac{29534}{(d_c)^2}\text{N/mm}^2$$

$$\sigma_v = \frac{P_v}{A_c}$$

$$= \frac{4220}{(d_c)^2}\text{N/mm}^2$$

5. Soderberg Fatigue Failure Equation

Soderberg line equation substitution

$$\sigma_v = \left(\frac{\sigma_e}{\text{F.S.}}\right)\left[1 - \frac{\sigma_m \cdot \text{F.S.}}{\sigma_y}\right]$$

$$\frac{4220}{(d_c)^2} = \left(\frac{240}{2}\right)\left[1 - \left(\frac{29534}{(d_c)^2}\right) \cdot \left(\frac{2}{330}\right)\right]$$

$$d_c = 14.63 \text{ mm}$$

6. Standard Thread Selection (Table 11.1)

Select standard coarse thread

$$d_{\text{c, std}} \geq d_c$$

$$= 14.63 \text{ mm}$$

$$d_c = 14.933 \text{ mm}$$

$$d = 18 \text{ mm}$$

7. Design Output	**Select Fastener Designation: M 18 Bolt ($d = 18\ \mathrm{mm}, d_c = 14.933\ \mathrm{mm}$)**

SECTION 10: Boiler Longitudinal Bar Stay Sizing

System Parameters

Pitch of Stays: $350 \text{ mm} \times 350 \text{ mm}$

Steam Pressure: $p = 0.84\text{N/mm}^2$

Permissible Tensile Stress: $\sigma_t = 56 \text{ MPa} = 56\text{N/mm}^2$

Design Protocol

Calculate area supported by each stay $A = p_s \times p_s$

Calculate steam force $P = A \cdot p$

Solve required core diameter d_c and select bolt size

1. Area Supported by Each Stay (A)

Area of boiler plate supported per stay

$$A = p_s \times p_s$$
$$= 350 \times 350$$
$$= 122500\text{mm}^2$$

2. Steam Force Acting on Stay (P)

Total steam pressure load on supported area

$$P = A \cdot p$$
$$= 122500\text{mm}^2 \times 0.84\text{N/mm}^2$$
$$= 102900 \text{ N}$$

3. Required Core Root Diameter (d_c)

Core diameter required for permissible tension stress

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$102900 = \frac{\pi}{4}(d_c)^2 \times 56$$

$$d_c = 48.37 \text{ mm}$$

4. Standard Thread Selection (Table 11.1)

Select standard coarse thread

$$d_{\text{c, std}} \geq d_c$$

$$= 48.37 \text{ mm}$$

$$d_c = 49.177 \text{ mm}$$

$$d = 56 \text{ mm}$$

5. Design Output

Select Fastener Designation: M 56 Bolt ($d = 56$ mm, $d_c = 49.177$ mm)

SECTION 10: Boiler Bar Stay Arrangement — MECHANICAL DIAGRAM

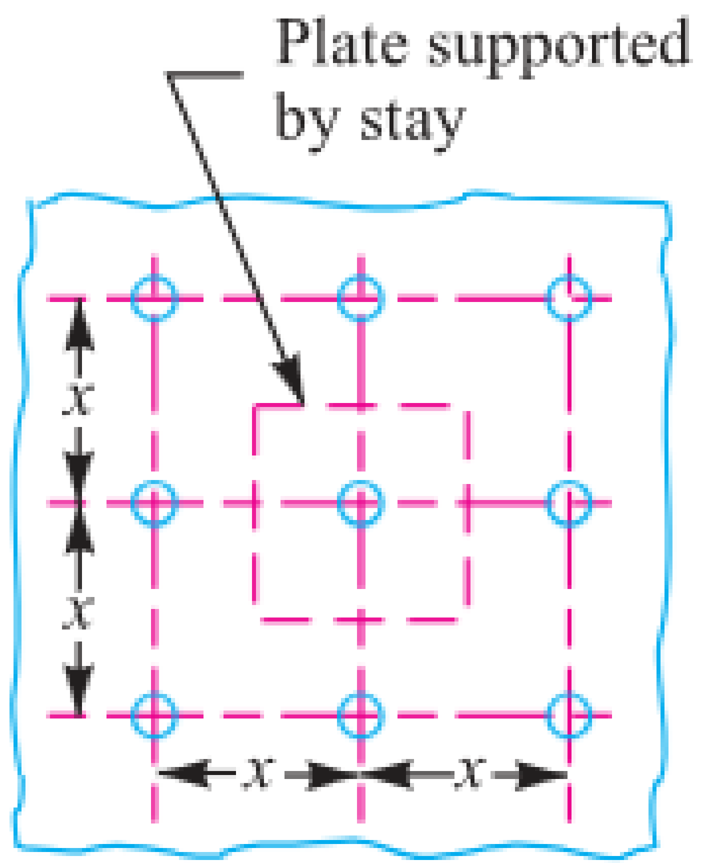


Fig. 11.29. Longitudinal bar stay.

Figure 11.29: Boiler Longitudinal Bar Stay Mounting Scheme

SECTION 11: Turned Shank & Drilled Hole Bolts of Uniform Strength

System Parameters

Nominal Bolt Size: $D_o = 48$ mm (M 48 Coarse Series)

Objective: Determine drilled axial hole diameter D for uniform strength

Design Protocol

Lookup core root diameter D_c from Table 11.1

Equate unthreaded drilled shank area to core root area

Solve $D = \sqrt{D_o^2 - D_c^2}$

1. Core Diameter Lookup (D_c)

Table 11.1 lookup for M 48 coarse thread

$$D_c = f(D_o)$$
$$= 41.795 \text{ mm}$$

2. Area Equality Condition for Uniform Strength

Equate unthreaded drilled shank area to root area

$$\frac{\pi}{4}\left(D_o^2 - D^2\right) = \frac{\pi}{4}D_c^2$$
$$D_o^2 - D^2 = D_c^2$$

3. Drilled Axial Hole Diameter (D)

Solve central axial hole diameter

$$D = \sqrt{D_o^2 - D_c^2}$$

$$= \sqrt{(48)^2 - (41.795)^2}$$

$$= \mathbf{23.64\ mm}$$

4. Design Output / Hole Diameter

Diameter of Hole $D = 23.64$ mm Ans.

SECTION 12: Wall Brackets under Parallel Eccentric Tensile Loading

System Parameters

Load: $W = 30 \text{ kN} = 30000 \text{ N}$

Eccentricity Arm: $L = 500 \text{ mm}$

Distances: $L_1 = 80 \text{ mm}$, $L_2 = 250 \text{ mm}$

Bolt Count: $n = 4$ (2 per row) | Allowable Stress: $\sigma_t = 60 \text{ MPa}$

Design Protocol

Calculate direct tensile load $W_{\text{t1}} = \frac{W}{n}$

Calculate unit secondary load $w = \frac{WL}{2(L_1^2+L_2^2)}$

Calculate total tension $W_t = W_{\text{t1}} + wL_2$ and core size d_c

---

1. Direct Tensile Load per Bolt (W_{t1})

Direct load shared equally by 4 bolts

$$W_{\text{t1}} = \frac{W}{n}$$

$$= \frac{30000}{4}$$

$$= 7500 \text{ N}$$

$$= 7.5 \text{ kN}$$

2. Secondary Tensile Load on Upper Bolts (W_{t2})

Moment equilibrium load factor w

$$w = \frac{W \cdot L}{2(L_1^2 + L_2^2)}$$

$$= \frac{30 \times 500}{2(80^2 + 250^2)}$$

$$= 0.10886 \text{ kN/mm}$$

$$W_{t2} = w \cdot L_2$$

$$= 0.10886 \times 250$$

$$= 27.215 \text{ kN}$$

$$= 27215 \text{ N}$$

3. Total Maximum Tensile Load (W_t)

Combined direct and secondary tension on critical top bolts

$$W_t = W_{t1} + W_{t2}$$

$$= 7.5 + 27.215$$

$$= 34.715 \text{ kN}$$

$$= 34715 \text{ N}$$

4. Required Core Diameter (d_c)

Solve root diameter for top row fasteners

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$34715 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$d_c = 27.14 \text{ mm}$$

5. Standard Thread Selection (Table 11.1)

Select standard coarse series thread

$$d_{\text{c, std}} \geq d_c$$
$$= 27.14 \text{ mm}$$
$$d_c = 28.706 \text{ mm}$$
$$d = 33 \text{ mm}$$

6. Design Output

Select Fastener Designation: M 33 Bolt ($d = 33$ mm, $d_c = 28.706$ mm)

SECTION 12: Wall Bracket (Parallel Load) — MECHANICAL DIAGRAM

11.19 Eccentric Load Acting Parallel to the Axis of Bolts

Consider a bracket having a rectangular base bolted to a wall by means of four bolts as shown in Fig. 11.31. A little consideration will show that each bolt is subjected to a direct tensile load of

$W_{t1} = \frac{W}{n}$, where n is the number of bolts.

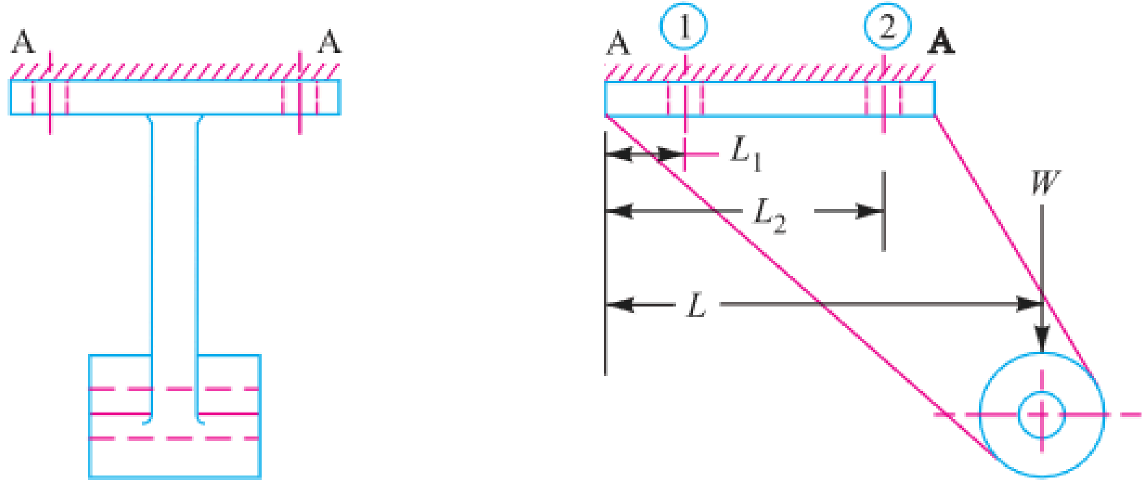


Fig. 11.31. Eccentric load acting parallel to the axis of bolts.

Figure 11.31: Wall Bracket under Parallel Eccentric Load

SECTION 13: Crane Runway T-Bracket Stress Analysis & Fastening Bolts

System Parameters

Wheel Load: $W = 15\ \mathrm{kN} = 15000\ \mathrm{N}$

Fastening Bolts: $d = 25\ \mathrm{mm}$ ($n = 4$)

T-Section: Flange 135×25, Web 175×25, Total Depth $200\ \mathrm{mm}$

Design Protocol

Part 1: Calculate centroid $\overline{y}$, moment of inertia I_{GG}, and max arm stresses σ_t, σ_c

Part 2: Calculate direct & secondary bolt tension to find bolt stress σ_{tb}

---

1. Centroid ($\overline{y}$) & Section Area (A)

T-section geometry evaluation

$$A_1 = 135 \times 25 = 3375 \mathrm{mm}^2$$

$$A_2 = 175 \times 25 = 4375 \mathrm{mm}^2$$

$$A = A_1 + A_2$$

$$= 7750 \mathrm{mm}^2$$

$$\overline{y} = \frac{A_1 y_1 + A_2 y_2}{A}$$

$$= \frac{3375 \times 12.5 + 4375 \times 112.5}{7750}$$

$$= 69\ \mathrm{mm} \quad (\mathrm{Top}\ \ y_1 = 69\ \mathrm{mm}, \mathrm{Bottom}\ \ y_2 = 131\ \mathrm{mm})$$

2. Moment of Inertia (I_{GG}) & Section Moduli (Z_1, Z_2)

Parallel axis theorem for T-section

$$I_{\mathrm{GG}} = \sum\left(I_i + A_i d_i^2\right)$$

$$= \left[\frac{135(25)^3}{12} + 3375(56.5)^2\right] + \left[\frac{25(175)^3}{12} + 4375(43.5)^2\right]$$

$$= 30.4 \times 10^6 \mathrm{mm}^4$$

$$Z_1 = \frac{I_{\mathrm{GG}}}{y_1} = \frac{30.4 \times 10^6}{69} = 440.6 \times 10^3 \mathrm{mm}^3$$

$$Z_2 = \frac{I_{\mathrm{GG}}}{y_2} = \frac{30.4 \times 10^6}{131} = 232 \times 10^3 \mathrm{mm}^3$$

3. Bending Moment at Section X-X (M)

Moment exerted on bracket arm section

$$M = W\left(200 + \overline{y}\right)$$

$$= 15 \times 10^3 (200 + 69)$$

$$= 4035 \times 10^3 \ \mathrm{N \cdot mm}$$

4. Maximum Tensile Stress at Section X-X (σ_t)

Bending tensile plus direct tensile stress in flange

$$\sigma_t = \frac{M}{Z_1} + \frac{W}{A}$$

$$= \frac{4035 \times 10^3}{440.6 \times 10^3} + \frac{15000}{7750}$$

$$= 9.16 + 1.94$$

$$= \mathbf{11.1\ MPa} \quad (\mathbf{Ans.})$$

5. Maximum Compressive Stress at Section X-X (σ_c)

Bending compressive minus direct tensile stress in web

$$\sigma_c = \frac{M}{Z_2} - \frac{W}{A}$$
$$= \frac{4035 \times 10^3}{232 \times 10^3} - 1.94$$
$$= 17.4 - 1.94$$
$$= \mathbf{15.46\ MPa} \quad (\mathbf{Ans.})$$

6. Direct Tensile Load per Bolt (W_{t1})

Direct shear/tension shared equally by 4 bolts

$$W_{t1} = \frac{W}{n}$$
$$= \frac{15000}{4}$$
$$= 3750\ \text{N}$$

7. Secondary Load per Unit Distance (w) & Load (W_{t2})

Secondary tension on critical upper bolts 2 & 3

$$w = \frac{W \cdot L}{2\left(L_1^2 + L_2^2\right)}$$
$$= \frac{15000 \times 525}{2(50^2 + 375^2)}$$
$$= 27.5\ \text{N/mm}$$
$$W_{t2} = w \cdot L_2 = 27.5 \times 375 = 10312\ \text{N}$$

8. Total Tensile Load on Critical Bolts (W_t)

Combined direct and secondary load on bolts 2 & 3

$$W_t = W_{\text{t1}} + W_{\text{t2}}$$
$$= 3750 + 10312$$
$$= 14062 \text{ N}$$

9. Maximum Tensile Stress in Fastening Bolts (σ_{tb})

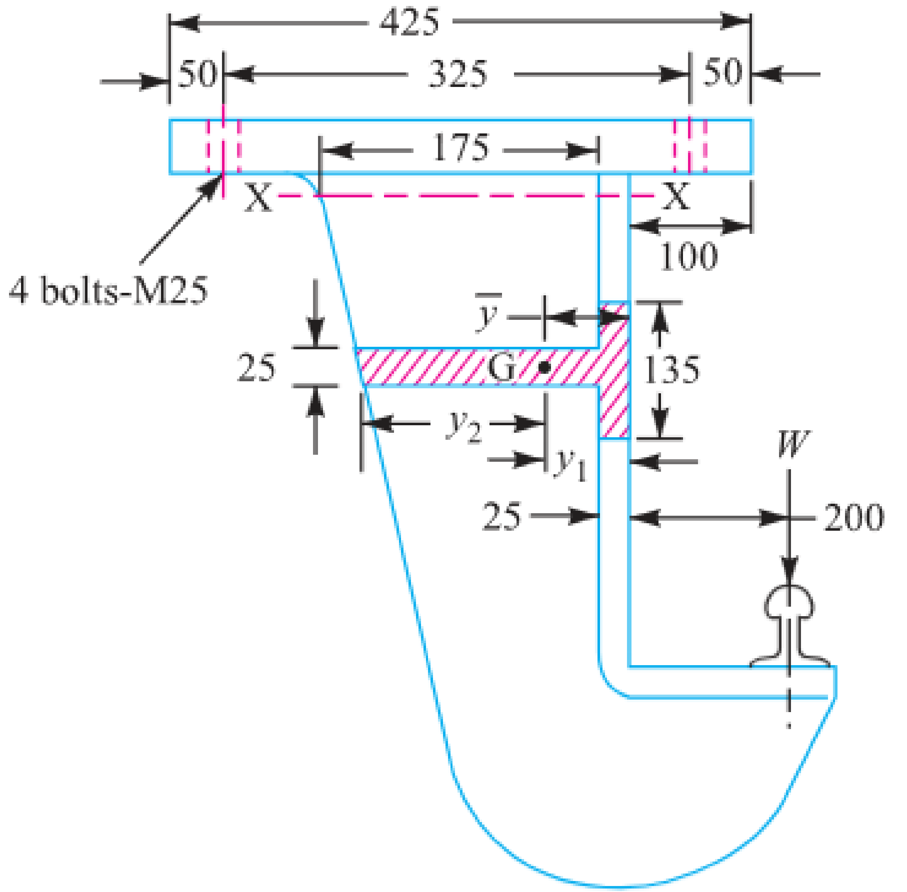
Core diameter stress evaluation ($d_c = 0.84d = 21$ mm)

$$\sigma_{\text{tb}} = \frac{W_t}{\frac{\pi}{4}(d_c)^2}$$
$$= \frac{14062}{\frac{\pi}{4}(21)^2}$$
$$= \frac{14062}{346.4}$$
$$= \mathbf{40.6\ MPa} \quad (\mathbf{Ans.})$$

10. Design Output

Arm Tensile $\sigma_t = 11.1$ MPa, Arm Compressive $\sigma_c = 15.46$ MPa, Bolt Stress $\sigma_{\text{tb}} = 40.6$ MPa

SECTION 13: Crane Runway T-Bracket Structure — MECHANICAL DIAGRAM



All dimensions in mm

Fig. 11.32

Figure 11.32: Crane Runway T-Bracket Structure

SECTION 13: Crane Runway T-Bracket Stress Distribution — MECHANICAL DIAGRAM

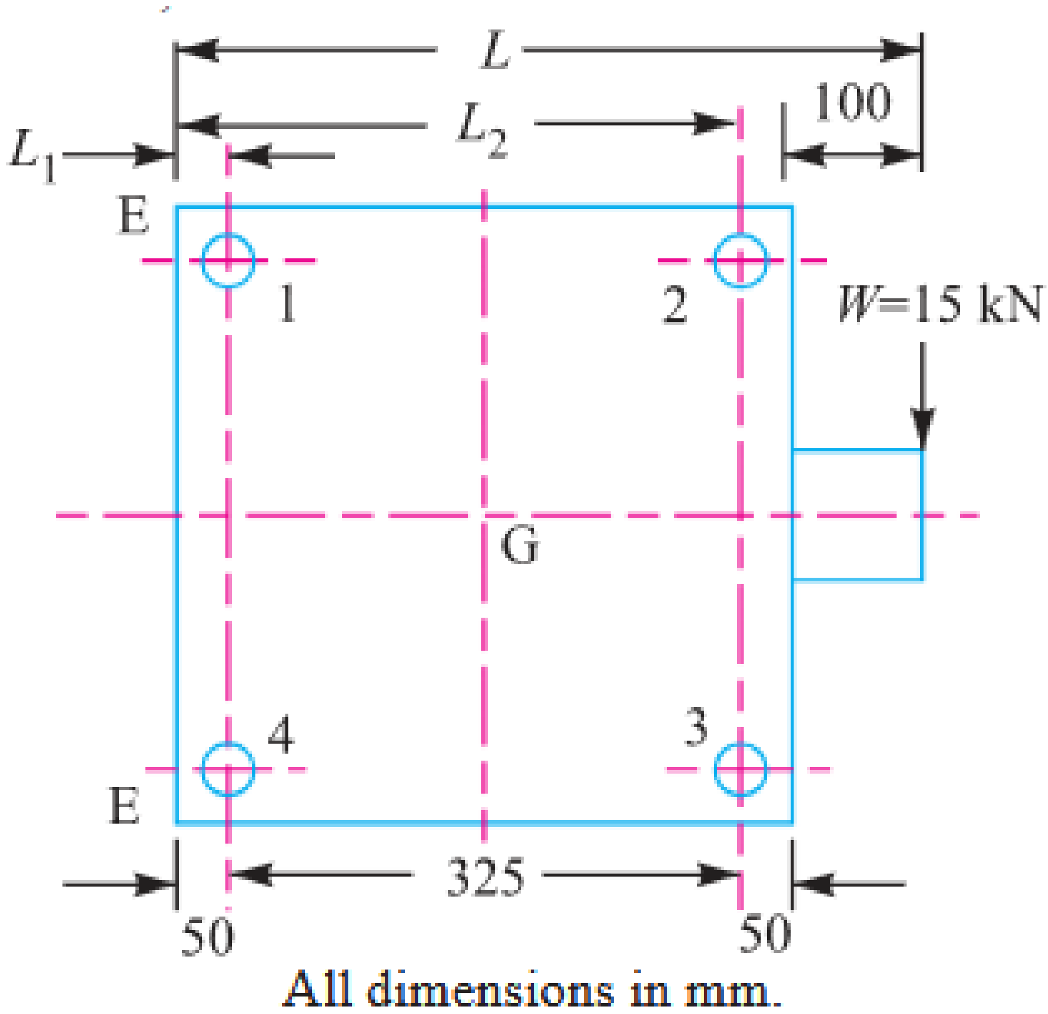


Fig. 11.33

Figure 11.33: T-Section Stress Distribution Diagram

SECTION 14: Travelling Crane Bracket Sizing (Combined Shear & Tension)

System Parameters

Load: $W = 12 \text{ kN} = 12000 \text{ N}$

Load Arm: $L = 400 \text{ mm}$

Distances: $L_1 = 50 \text{ mm}$, $L_2 = 375 \text{ mm}$

Permissible Tensile Stress: $\sigma_t = 84 \text{ MPa}$ | Bolt Count: $n = 4$

Design Protocol

Part 1: Calculate direct shear W_s, secondary tension W_t, and equivalent load W_{te} via Maximum Principal Stress Theory

Part 2: Calculate arm thickness t for depth $b = 250 \text{ mm}$

---

1. Direct Shear Load per Bolt (W_s)

Direct shear force carried per bolt

$$W_s = \frac{W}{n}$$

$$= \frac{12}{4}$$

$$= 3 \text{ kN}$$

2. Maximum Secondary Tensile Load on Upper Bolts (W_t)

Tilting tension on critical upper bolts 3 & 4

$$W_t = \frac{W \cdot L \cdot L_2}{2\left(L_1^2 + L_2^2\right)}$$
$$= \frac{12 \times 400 \times 375}{2(50^2 + 375^2)}$$
$$= 6.29 \text{ kN}$$

3. Equivalent Tensile Load (W_{te})

Maximum Principal Stress Theory for combined shear & tension

$$W_{\text{te}} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$
$$= \frac{1}{2}\left[6.29 + \sqrt{(6.29)^2 + 4(3)^2}\right]$$
$$= \frac{1}{2}[6.29 + 8.69]$$
$$= 7.49 \text{ kN}$$
$$= 7490 \text{ N}$$

4. Required Core Root Diameter (d_c)

Tensile stress capacity equation

$$W_{\text{te}} = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$7490 = \frac{\pi}{4}(d_c)^2 \times 84$$

$$= 66(d_c)^2$$

$$(d_c)^2 = \frac{7490}{66} = 113.5$$

$$d_c = 10.65 \text{ mm}$$

5. Standard Fastener Selection (Table 11.1)

Select standard metric coarse thread

Select Fastener Designation: M 14 Bolt ($d_c = 11.546 \text{ mm} > 10.65 \text{ mm}$) quad bold("(Ans.)")

6. Maximum Bending Moment on Bracket Arm (M)

Moment about column face

$$M = W \cdot L$$

$$= 12 \times 10^3 \times 400$$

$$= 4.8 \times 10^6 \text{ N} \cdot \text{mm}$$

7. Arm Section Modulus & Dimension Relation ($t \cdot b^2$)

Flexural stress equation for rectangular section ($Z = \frac{1}{6}tb^2$)

$$\sigma_t = \frac{M}{Z}$$

$$84 = \frac{4.8 \times 10^6}{\frac{1}{6} \cdot t \cdot b^2} = \frac{28.8 \times 10^6}{t \cdot b^2}$$

$$t \cdot b^2 = \frac{28.8 \times 10^6}{84}$$

$$= 343 \times 10^3 \text{mm}^3$$

8. Bracket Arm Thickness (t)

Assuming standard bracket depth $b = 250$ mm

$$t = \frac{343 \times 10^3}{b^2}$$

$$= \frac{343 \times 10^3}{(250)^2}$$

$$= \mathbf{5.5\ mm} \quad (\mathbf{Ans.})$$

9. Design Output

Fasteners: M 14 Bolts, Bracket Arm: $t = 5.5$ mm for depth $b = 250$ mm

SECTION 14: Travelling Crane Bracket — MECHANICAL DIAGRAM

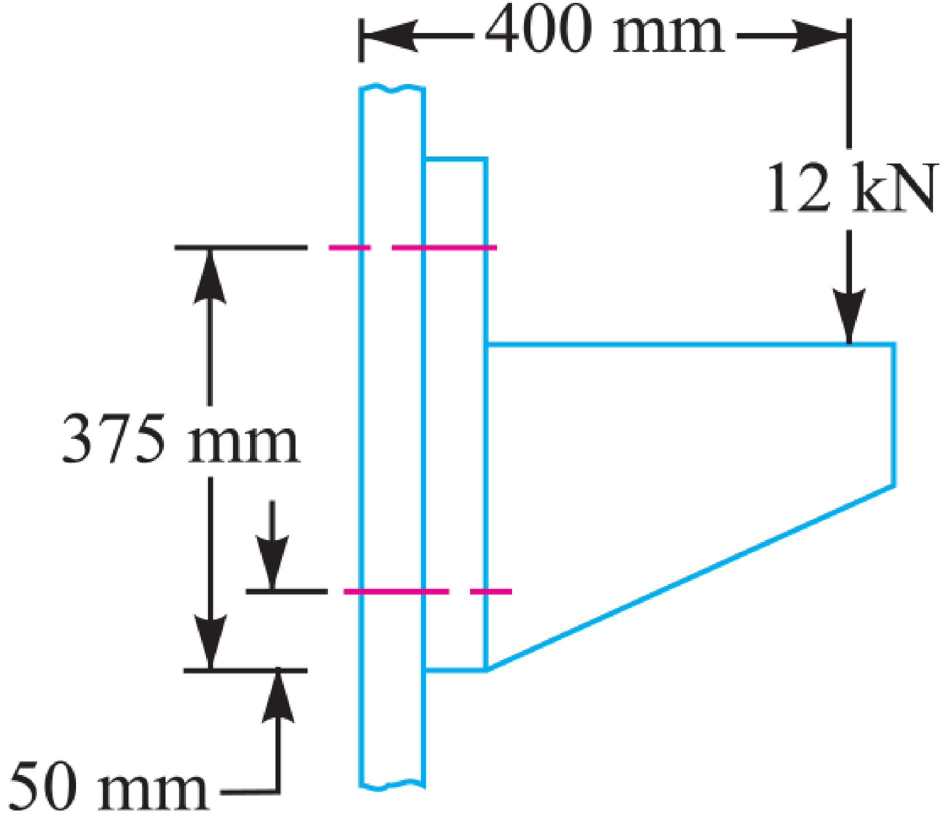


Fig. 11.35

Figure 11.35: Travelling Crane Wall Bracket Structural Geometry

SECTION 15: Inclined Load Bracket Fastener Sizing & Arm Thickness Analysis

System Parameters

Inclined Load: $W = 40 \text{ kN} = 40000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical

Stresses: $\sigma_t = 70 \text{ MPa}, \tau = 50 \text{ MPa}, \sigma_c = 105 \text{ MPa}$

Arm Depth: $b = 130 \text{ mm}$ | Bolts: $n = 4, L_1 = 60 \text{ mm}, L_2 = 180 \text{ mm}$

Design Protocol

Part 1: Resolve load into W_H, W_V and net moment $T_{\text{net}} = T_V - T_H$

Part 2: Calculate bolt tension W_t, equivalent load W_{te}, and size d_c

Part 3: Calculate arm thickness t via principal tensile stress $\sigma_{\text{t(max)}}$

1. Force Component Resolution (W_H, W_V)

Horizontal and vertical force components

$$W_H = W \sin\theta$$

$$= 40 \sin 60^{\text{deg}} = 40 \times 0.866$$

$$= 34.64 \text{ kN} = 34640 \text{ N}$$

$$W_V = W \cos\theta$$

$$= 40 \cos 60^{\text{deg}} = 40 \times 0.5$$

$$= 20 \text{ kN} = 20000 \text{ N}$$

2. Net Turning Moment (T_{net})

Clockwise turning moment about tilting edge E

$$T_H = W_H \times 20 = 34640 \times 20$$
$$= 692.8 \times 10^3 \text{ N} \cdot \text{mm} \quad \text{(Anticlockwise)}$$
$$T_V = W_V \times 175 = 20000 \times 175$$
$$= 3500 \times 10^3 \text{ N} \cdot \text{mm} \quad \text{(Clockwise)}$$
$$T_{\text{net}} = T_V - T_H$$
$$= (3500 - 692.8) \times 10^3$$
$$= 2807.2 \times 10^3 \text{ N} \cdot \text{mm} \quad \text{(Clockwise)}$$

3. Direct Loads per Bolt (W_{t1}, W_s)

Direct tensile load and direct shear load per bolt

$$W_{\text{t1}} = \frac{W_H}{n} = \frac{34640}{4} = 8660 \text{ N}$$
$$W_s = \frac{W_V}{n} = \frac{20000}{4} = 5000 \text{ N}$$

4. Secondary Tensile Load on Upper Bolts (W_{t2})

Secondary load per unit distance w and upper bolt load

$$w = \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)} = \frac{2807.2 \times 10^3}{2(60^2 + 180^2)}$$
$$= \frac{2807.2 \times 10^3}{72000} = 39 \text{ N/mm}$$
$$W_{\text{t2}} = w \cdot L_2 = 39 \times 180 = 7020 \text{ N}$$

5. Total Tensile & Equivalent Tensile Load (W_t, W_{te})

Maximum Principal Stress Theory combination

$$W_t = W_{t1} + W_{t2} = 8660 + 7020 = 15680 \text{ N}$$

$$W_{\text{te}} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$

$$= \frac{1}{2}\left[15680 + \sqrt{(15680)^2 + 4(5000)^2}\right]$$

$$= \frac{1}{2}[15680 + 18600]$$

$$= 17140 \text{ N}$$

6. Required Core Root Diameter (d_c) & Fastener Selection

Table 11.1 coarse series bolt selection

$$17140 = \frac{\pi}{4}(d_c)^2 \times 70 = 55(d_c)^2$$

$$(d_c)^2 = \frac{17140}{55} = 311.64$$

$$d_c = 17.65 \text{ mm}$$

[Select Fastener Designation: M 22 Bolt ($d_c = 18.933 \text{ mm} > 17.65 \text{ mm}$) quad bold(“(Ans.)”)]

7. Arm Section Area (A) & Section Modulus (Z)

Bracket arm cross-sectional properties for depth $b = 130$ mm

$$A = b \cdot t = 130t\text{mm}^2$$

$$Z = \frac{1}{6}tb^2 = \frac{1}{6}t(130)^2 = 2817t\text{mm}^3$$

8. Tensile and Shear Stresses in Arm (σ_t, τ)

Direct and bending stress components in upper fibre

$$\sigma_{\mathrm{t1}} = \frac{W_H}{A} = \frac{34640}{130t} = \frac{266.5}{t}$$

$$\sigma_{\mathrm{t2}} = \frac{M_H}{Z} = \frac{34640 \times 35}{2817t} = \frac{430.4}{t}$$

$$\sigma_{\mathrm{t3}} = \frac{M_V}{Z} = \frac{20000 \times 200}{2817t} = \frac{1420}{t}$$

$$\sigma_t = \sigma_{\mathrm{t1}} + \sigma_{\mathrm{t2}} + \sigma_{\mathrm{t3}} = \frac{266.5 + 430.4 + 1420}{t} = \frac{2116.9}{t}$$

$$\tau = \frac{W_V}{A} = \frac{20000}{130t} = \frac{154}{t}$$

9. Maximum Principal Tensile Stress & Arm Thickness (t)

Thickness evaluation for allowable tensile stress $\sigma_t = 70\ \mathrm{MPa}$

$$\sigma_{\mathrm{t(max)}} = \frac{\sigma_t}{2} + \frac{1}{2}\sqrt{\sigma_t^2 + 4\tau^2}$$

$$= \frac{1058.45}{t} + \frac{1069.6}{t} = \frac{2128.05}{t}$$

$$70 = \frac{2128.05}{t}$$

$$t = \frac{2128.05}{70} = 30.4\ \mathrm{mm} \qquad \Rightarrow \textbf{Adopt Arm Thickne}$$

10. Induced Shear Stress Verification ($\tau_{\max}$)

Maximum shear stress check for $t = 31$ mm

$$\tau_{\max} = \frac{1069.6}{t} = \frac{1069.6}{31} = 34.5 \text{ MPa} \leq 50 \text{ MPa} \quad (\textbf{SAFE})$$

11. Lower Fibre Compressive Stress Check (σ_c)

Check arm thickness against allowable compressive stress $\sigma_c = 105$ MPa

$$\sigma_c = -\sigma_{\mathrm{t1}} + \sigma_{\mathrm{t2}} + \sigma_{\mathrm{t3}} = \frac{-266.5 + 430.4 + 1420}{t} = \frac{1583.9}{t}$$

$$105 = \frac{1583.9}{t}$$

$$t = 15.1 \text{ mm} \quad (\textbf{Governed by Tensile Limit } \ t = 31 \text{ mm})$$

12. Design Output

Final Fasteners: M 22 Bolts, Bracket Arm Dimensions $t = 31$ mm for depth $b = 130$ mm

SECTION 15: Inclined Load Bracket Structure — MECHANICAL DIAGRAM

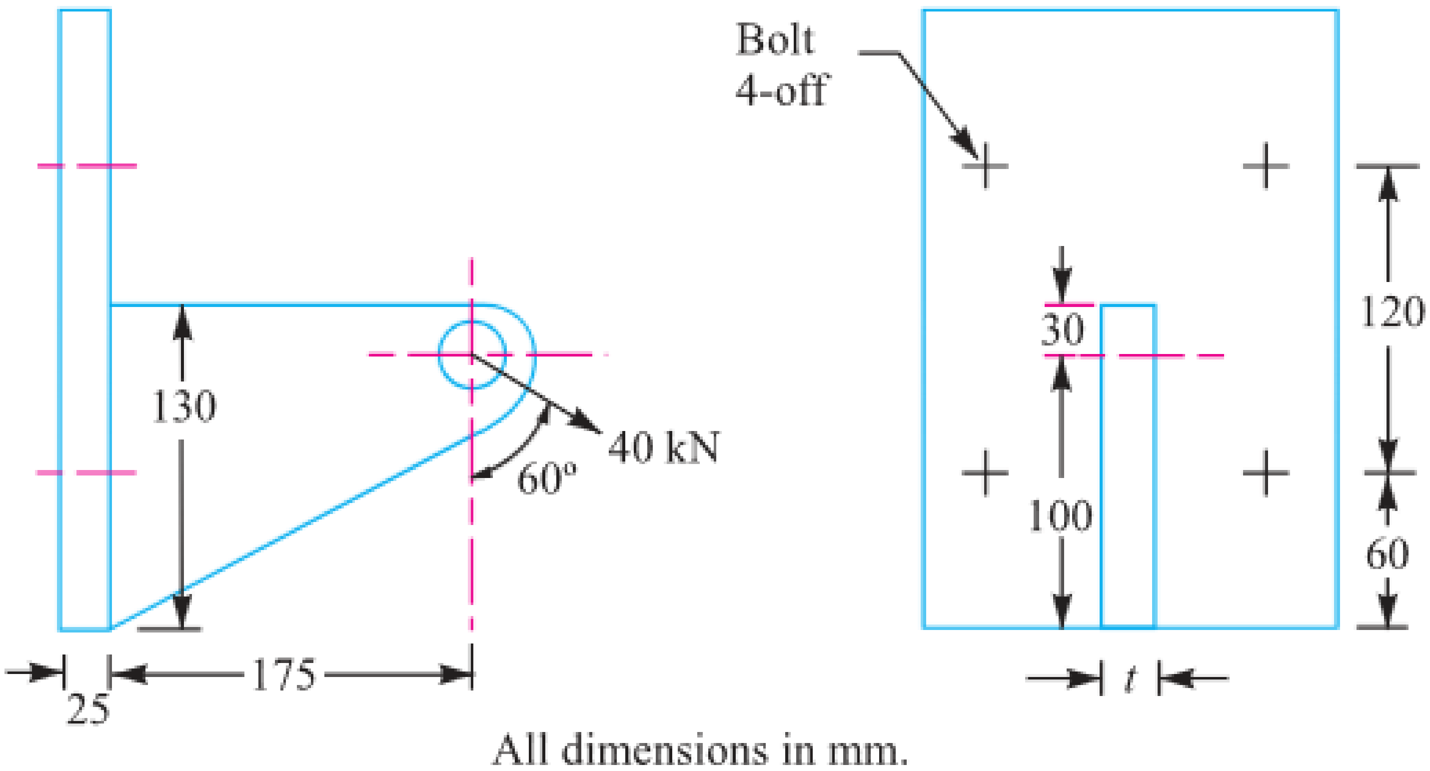
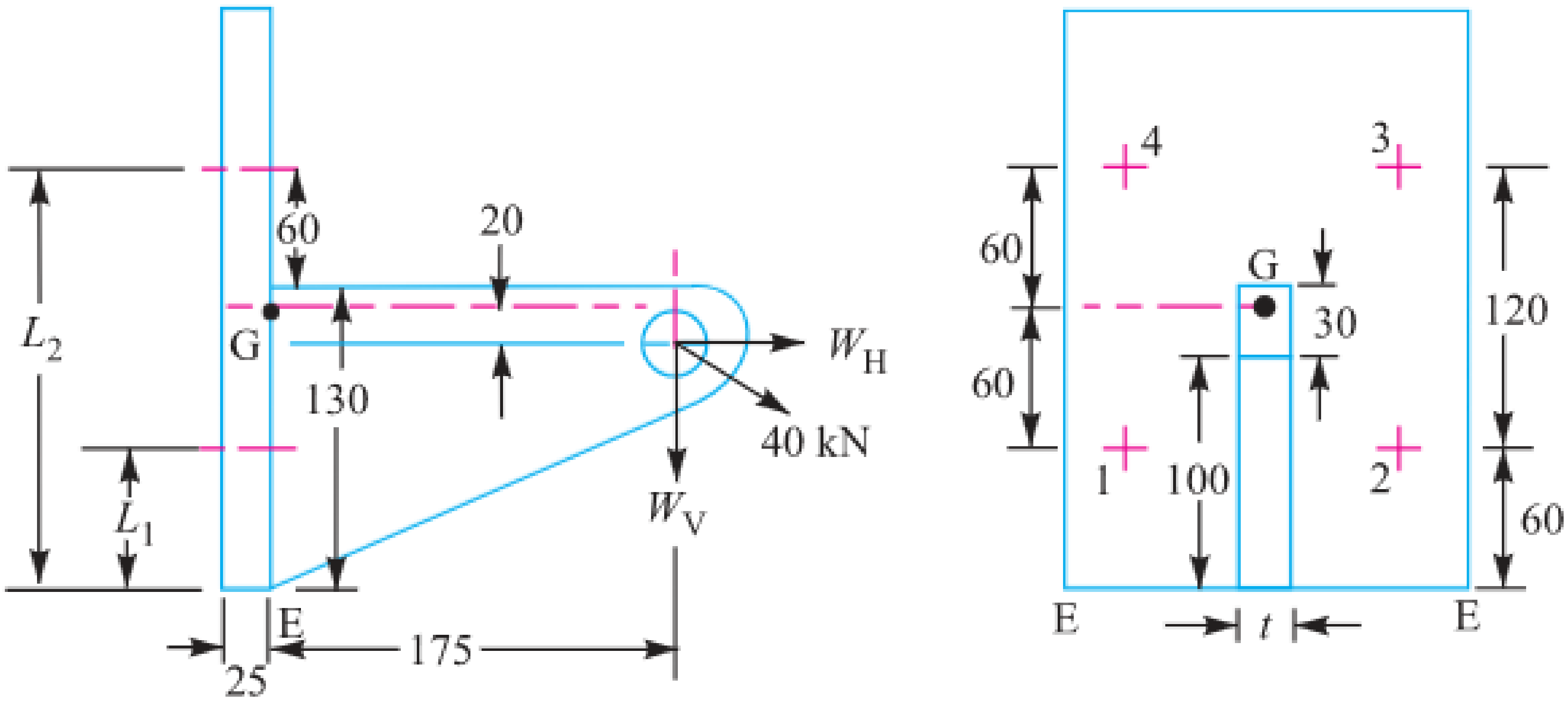


Fig 11.36

Figure 11.36: Wall Bracket under Inclined Load

SECTION 15: Inclined Load Bracket Force Resolution — MECHANICAL DIAGRAM



All dimensions in mm.

Fig. 11.37

Figure 11.37: Load Resolution & Moment Arm Diagram

SECTION 16: Offset I-Section Bracket under Inclined Loading

System Parameters

Pull Load: $W = 10 \text{ kN} = 10000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical

Stresses: $\sigma_t = 100 \text{ MPa}, \tau = 60 \text{ MPa}$

Bolt Count: $n = 4$ | I-Section Flange Width: $b = 3t$

Design Protocol

Part 1: Resolve load into W_H, W_V and net moment $T_{\text{net}} = T_V - T_H$

Part 2: Calculate bolt tension W_t, equivalent load W_{te}, and size d_c

Part 3: Calculate I-section arm thickness t and width $b = 3t$

---

1. Force Component Resolution (W_H, W_V)

Horizontal pull and vertical load components

$$W_H = W \sin\theta$$

$$= 10 \sin 60^{\text{deg}} = 10 \times 0.866$$

$$= 8.66 \text{ kN} = 8660 \text{ N}$$

$$W_V = W \cos\theta$$

$$= 10 \cos 60^{\text{deg}} = 10 \times 0.5$$

$$= 5 \text{ kN} = 5000 \text{ N}$$

2. Net Turning Moment (T_{net})

Counterclockwise turning moment about tilting edge E

$$T_H = W_H \times 0.05 = 8660 \times 0.05$$
$$= 433 \text{ N} \cdot \text{m} \quad \text{(Clockwise)}$$
$$T_V = W_V \times 0.3 = 5000 \times 0.3$$
$$= 1500 \text{ N} \cdot \text{m} \quad \text{(Anticlockwise)}$$
$$T_{\text{net}} = T_V - T_H$$
$$= 1500 - 433$$
$$= 1067 \text{ N} \cdot \text{m} \quad \text{(Anticlockwise)}$$

3. Direct Loads per Bolt (W_{t1}, W_s)

Direct tensile load and direct shear load per bolt

$$W_{\text{t1}} = \frac{W_H}{n} = \frac{8660}{4} = 2165 \text{ N}$$
$$W_s = \frac{W_V}{n} = \frac{5000}{4} = 1250 \text{ N}$$

4. Secondary Tensile Load on Upper Bolts (W_{t2})

Tilting edge distances $L_1 = 0.0375$ m, $L_2 = 0.2125$ m

$$w = \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)} = \frac{1067}{2(0.0375^2 + 0.2125^2)}$$
$$= \frac{1067}{0.093} = 11470 \text{ N/m}$$
$$W_{\text{t2}} = w \cdot L_2 = 11470 \times 0.2125 = 2435 \text{ N}$$

5. Total Tensile & Equivalent Tensile Load (W_t, W_{te})

Maximum Principal Stress Theory combination

$$W_t = W_{t1} + W_{t2} = 2165 + 2435 = 4600 \text{ N}$$

$$W_{te} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$

$$= \frac{1}{2}\left[4600 + \sqrt{(4600)^2 + 4(1250)^2}\right]$$

$$= \frac{1}{2}[4600 + 5240]$$

$$= 4920 \text{ N}$$

6. Required Core Root Diameter (d_c) & Fastener Selection

Table 11.1 coarse series bolt selection

$$4920 = \frac{\pi}{4}(d_c)^2 \times 100 = 78.55(d_c)^2$$

$$(d_c)^2 = \frac{4920}{78.55} = 62.6$$

$$d_c = 7.91 \text{ mm}$$

[Select Fastener Designation: M 10 Bolt ($d_c = 8.18$ mm $>$ 7.91 mm) quad bold("(Ans.)")]

7. I-Section Arm Section Area (A) & Modulus (Z)

Geometrical properties for flange width $b = 3t$

$$A = 3b \cdot t = 3(3t)t = 9t^2\text{mm}^2$$

$$Z = \frac{I}{1.5t} = \frac{321\frac{t^4}{12}}{1.5t} = 10.7t^3\text{mm}^3$$

8. Tensile and Bending Stresses in Arm ($\sigma_{t1}, \sigma_{t2}, \sigma_{t3}$)

Direct and bending stress components in top flange

$$\sigma_{t1} = \frac{W_H}{A} = \frac{8660}{9t^2} = \frac{962}{t^2}$$

$$\sigma_{t2} = \frac{M_H}{Z} = \frac{433 \times 10^3}{10.7t^3} = \frac{40.5 \times 10^3}{t^3}$$

$$\sigma_{t3} = \frac{M_V}{Z} = \frac{1500 \times 10^3}{10.7t^3} = \frac{140.2 \times 10^3}{t^3}$$

9. Net Tensile Stress & Arm Dimensions (t, b)

Hit-and-trial solution for allowable tensile stress $\sigma_t = 100$ MPa

$$\sigma_t = \sigma_{t1} - \sigma_{t2} + \sigma_{t3} = \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3}$$

$$100 = \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3}$$

$$t = 10.4 \text{ mm} \qquad \Rightarrow \textbf{Adopt Arm Thick}$$

$$b = 3t = 3 \times 10.4 = \mathbf{31.2\ mm} \quad \textbf{(Ans.)}$$

10. Design Output

Final Fasteners: M 10 Bolts, I-Section Arm Dimensions $t = 10.4$ mm, $b = 31.2$ mm

SECTION 16: Offset I-Section Bracket Structure — MECHANICAL DIAGRAM

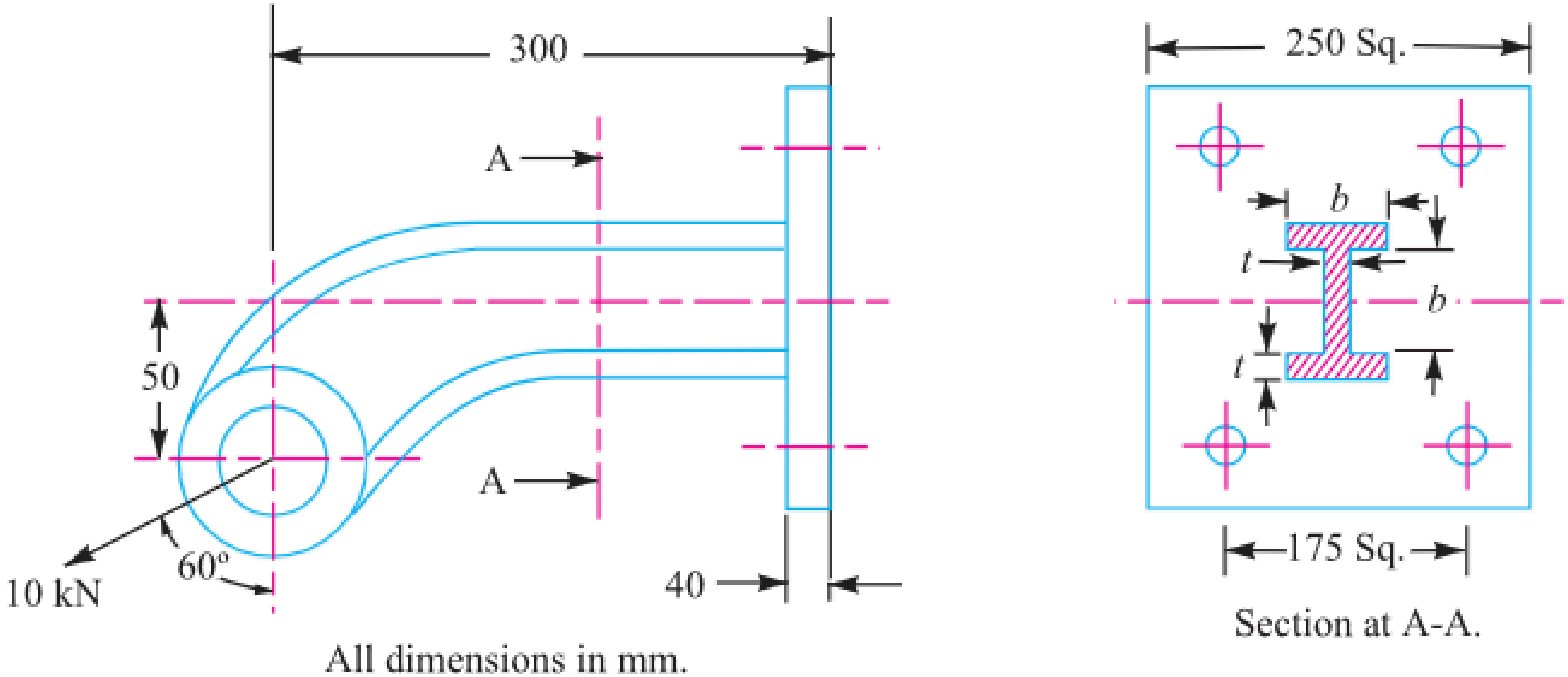


Fig. 11.38

Figure 11.38: Offset Bracket with I-Section Arm

SECTION 16: Offset I-Section Bracket Load Distribution — MECHANICAL DIAGRAM

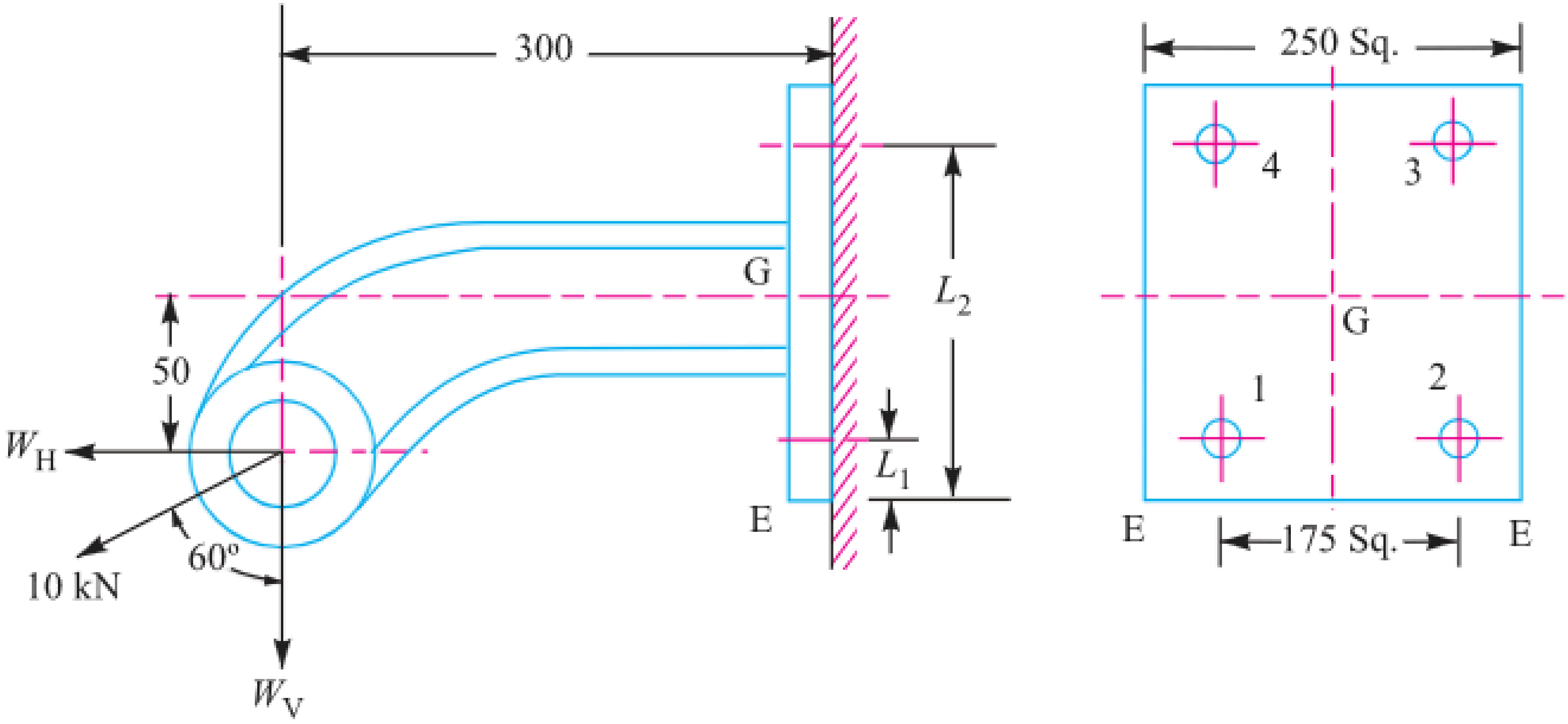


Fig. 11.39

Figure 11.39: Load Distribution and Force Moment Diagram

SECTION 17: Pillar Crane Circular Base Foundation Flange Sizing

System Parameters

Bolt Count: $n = 8$

Bolt Circle Diameter: $d = 1.6\ \mathrm{m} \Rightarrow r = 0.8\ \mathrm{m}$

Pillar Base Diameter: $D = 2\ \mathrm{m} \Rightarrow R = 1\ \mathrm{m}$

Crane Load: $W = 100\ \mathrm{kN} = 100000\ \mathrm{N}$ at distance $e = 5\ \mathrm{m}$ | Allowable Stress: $\sigma_t = 100\ \mathrm{MPa}$

Design Protocol

Calculate distance from tilting edge $A - A$: $L = e - R$

Calculate maximum tensile load W_t on critical bolt using PCD formula

Solve required core root diameter d_c and select bolt size

---

1. Tilting Arm Distance (L)

Distance measured from base fulcrum edge $A - A$ of radius $R = 1\ \mathrm{m}$

$$L = e - R$$
$$= 5 - 1$$
$$= 4\ \mathrm{m}$$

2. Maximum Tensile Load on Critical Bolt (W_t)

Circular foundation bolt layout equation for tilting about base edge

$$W_t = \frac{2 \cdot W \cdot L \cdot (R + r)}{n(2R^2 + r^2)}$$

$$= \frac{2 \times (100000) \times 4 \times (1 + 0.8)}{8 \times (2(1)^2 + 0.8^2)}$$

$$= 68180 \text{ N}$$

3. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$68180 = \frac{\pi}{4}(d_c)^2 \times 100$$

$$d_c = 29.46 \text{ mm}$$

4. Standard Thread Selection (Table 11.1)

Select standard coarse thread

$$d_{\text{c, std}} \geq d_c$$

$$= 29.46 \text{ mm}$$

$$d_c = 31.093 \text{ mm}$$

$$d = 36 \text{ mm}$$

5. Design Output

Select Fastener Designation: M 36 Bolt ($d = 36$ mm, $d_c = 31.093$ mm)

SECTION 17: Pillar Crane Foundation Flange — MECHANICAL DIAGRAM

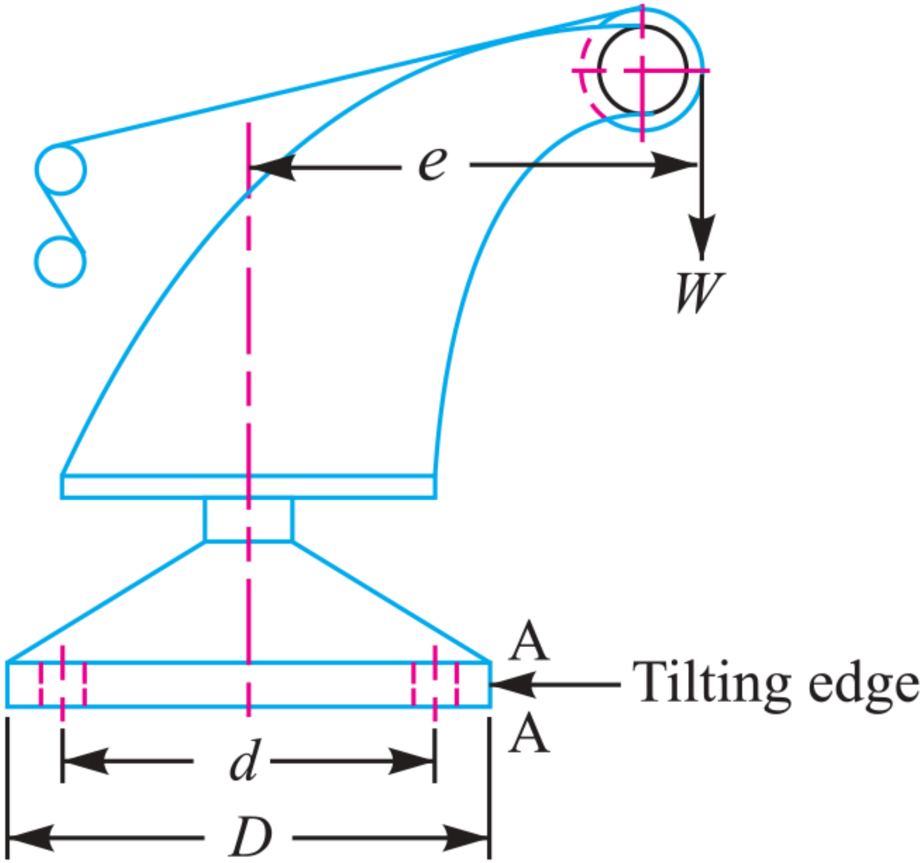


Fig. 11.42

Figure 11.42: Pillar Crane Base Flange 8-Bolt PCD Layout

SECTION 18: Circular Flanged Bearing Base Fastener Sizing

System Parameters

Bolt Count: $n = 4$

Bolt Circle Diameter: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm}$

Bearing Flange Diameter: $D = 650 \text{ mm} \Rightarrow R = 325 \text{ mm}$

Overturning Load: $W = 400 \text{ kN} = 400000 \text{ N}$ at arm $L = 250 \text{ mm}$ | Stress: $\sigma_t = 60 \text{ MPa}$

Design Protocol

For $n = 4$ bolts on circular flange, critical bolt lies at angle $\alpha = \frac{180^{\text{deg}}}{n} = 45^{\text{deg}}$

Apply circular PCD overturning equation for W_t

Solve required core root diameter d_c

---

1. Maximum Tensile Load on Critical Bolt (W_t)

Overturning equation for $n = 4$ bolts at $\alpha = 45^{\text{deg}}$

$$W_t = \frac{2 \cdot W \cdot L \cdot \left[R + r\cos\left(\frac{180^{\text{deg}}}{n}\right)\right]}{n(2R^2 + r^2)}$$

$$= \frac{2 \times (400000) \times 250 \times \left[325 + 250\cos 45^{\text{deg}}\right]}{4 \times (2(325)^2 + (250)^2)}$$

$$= 91643 \text{ N}$$

2. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$91643 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$d_c = 44.1 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse thread

$$d_{\text{c, std}} \geq d_c$$

$$= 44.1 \text{ mm}$$

$$d_c = 45.795 \text{ mm}$$

$$d = 52 \text{ mm}$$

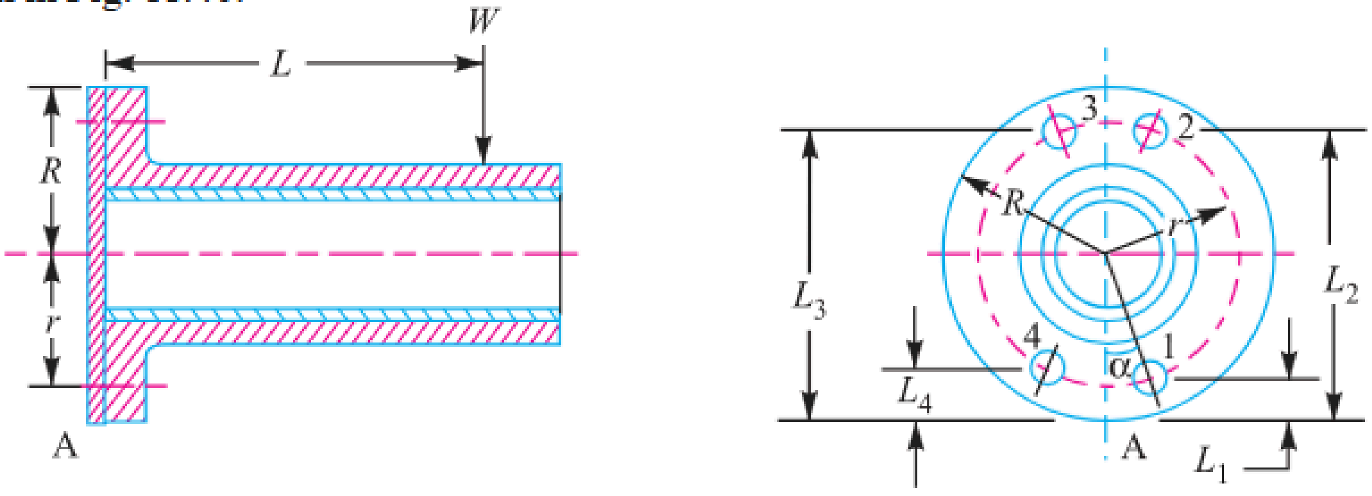
4. Design Output

Select Fastener Designation: M 52 Bolt ($d = 52$ mm, $d_c = 45.795$ mm)

SECTION 18: Flanged Bearing Base — MECHANICAL DIAGRAM

11.21 Eccentric Load on a Bracket with Circular Base

Sometimes the base of a bracket is made circular as in case of a flanged bearing of a heavy machine tool and pillar crane etc. Consider a round flange bearing of a machine tool having four bolts as shown in Fig. 11.40.



(a) (b)

Fig. 11.40. Eccentric load on a bracket with circular base.

Figure 11.40: Circular Flanged Bearing Base PCD Layout

SECTION 19: Pillar Crane Circular Foundation 2-Line Overturning Analysis

System Parameters

Base Diameter: $D = 600 \text{ mm} \Rightarrow R = 300 \text{ mm} = 0.3 \text{ m}$

PCD: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm} = 0.25 \text{ m}$

Fasteners: $n = 4$ M 30 bolts ($A_c = 561\text{mm}^2$)

Overturning Load: $W = 60 \text{ kN} = 60000 \text{ N}$ | Allowable Stress: $\sigma_t = 60 \text{ MPa}$

Design Protocol

Part 1: Analyze line X-X (45^{deg} to bolts) to find max distance e

Part 2: Analyze line Y-Y (in line with bolts) at same e to find max induced stress $\sigma_{\max}$

---

1. Bolt Tensile Capacity (P_{cap}) & Direct Load (W_{t1})

M 30 bolt capacity ($A_c = 561\text{mm}^2$) for allowable stress $\sigma_t = 60 \text{ MPa}$

$$P_{\text{cap}} = A_c \times \sigma_t = 561 \times 60 = 33660 \text{ N} = 33.66 \text{ kN}$$

$$W_{\text{t1}} = \frac{W}{n} = \frac{60}{4} = 15 \text{ kN}$$

2. Net Tensile Load on Critical Bolt ($P_{\max}$)

Sum of tensile capacity and direct load at distance L_2

$$P_{\max} = P_{\text{cap}} + W_{\text{t1}}$$
$$= 33.66 + 15$$
$$= 48.66 \text{ kN}$$

3. Line X-X Tilting Distances (L_1, L_2)
Bolt distances from tilting tangent axis A-A at 45^{deg} to bolts

$$L_1 = R - r\cos 45^{\text{deg}} = 0.3 - 0.25(0.7071) = 0.123 \text{ m}$$
$$L_2 = R + r\cos 45^{\text{deg}} = 0.3 + 0.25(0.7071) = 0.477 \text{ m}$$

---

4. Unit Load (w) & Resisting Moment about Axis A-A (M_{res})
Moment of bolt forces about outer tilting tangent A-A

$$w = \frac{P_{\max}}{L_2} = \frac{48.66}{0.477} = 102 \text{ kN/m}$$
$$M_{\text{res}} = 2w\left(L_1^2 + L_2^2\right)$$
$$= 2 \times 102 \times \left((0.123)^2 + (0.477)^2\right)$$
$$= 49.4 \text{ kN} \cdot \text{m}$$

---

5. Part 1: Maximum Load Distance (e)
Overturning moment equilibrium along line X-X

$$M_{\text{overturn}} = W(e - R) = 60(e - 0.3)$$
$$60(e - 0.3) = 49.4$$
$$e - 0.3 = \frac{49.4}{60} = 0.823 \text{ m}$$
$$e = 0.823 + 0.3 = \mathbf{1.123 \text{ m}} \quad (\textbf{Ans.})$$

---

6. Line Y-Y Tilting Distances (L_1, L_2, L_3)
Bolt distances from tilting tangent axis B-B

$$L_1 = R - r = 0.3 - 0.25 = 0.05 \text{ m} \quad (1 \text{ bolt})$$
$$L_2 = R = 0.3 \text{ m} \quad (2 \text{ bolts})$$
$$L_3 = R + r = 0.3 + 0.25 = 0.55 \text{ m} \quad (1 \text{ bolt, critical})$$

---

7. Resisting Moment & Unit Load along Line Y-Y

Resisting moment about axis B-B at $e = 1.123$ m

$$M_{\text{res}} = w(L_1^2 + 2L_2^2 + L_3^2)$$
$$= w((0.05)^2 + 2(0.3)^2 + (0.55)^2) = 0.485w$$

$$0.485w = 49.4 \Rightarrow w = \frac{49.4}{0.485} = 102 \text{ kN/m}$$

8. Tensile Load & Net Force on Critical Bolt (W_{t2}, P_{net})

Secondary load and net tensile load

$$W_{t2} = w \cdot L_3 = 102 \times 0.55 = 56.1 \text{ kN}$$

$$P_{\text{net}} = W_{t2} - W_{t1} = 56.1 - 15 = 41.1 \text{ kN} = 41100 \text{ N}$$

9. Part 2: Maximum Induced Tensile Stress (σ_{max})

Stress evaluation using textbook stress area $A_c = 516\text{mm}^2$

$$\sigma_{\max} = \frac{P_{\text{net}}}{A_c}$$
$$= \frac{41100}{516}$$
$$= \mathbf{79.65\ MPa} \quad \text{(Ans.)}$$

10. Design Output

Final Answers: (1) Maximum Load Distance $e = 1.123$ m, (2) Maximum Induced Stress $\sigma_{max} = 79.65$ MPa

SECTION 19: Pillar Crane 2-Line Overturning Analysis — MECHANICAL DIAGRAM

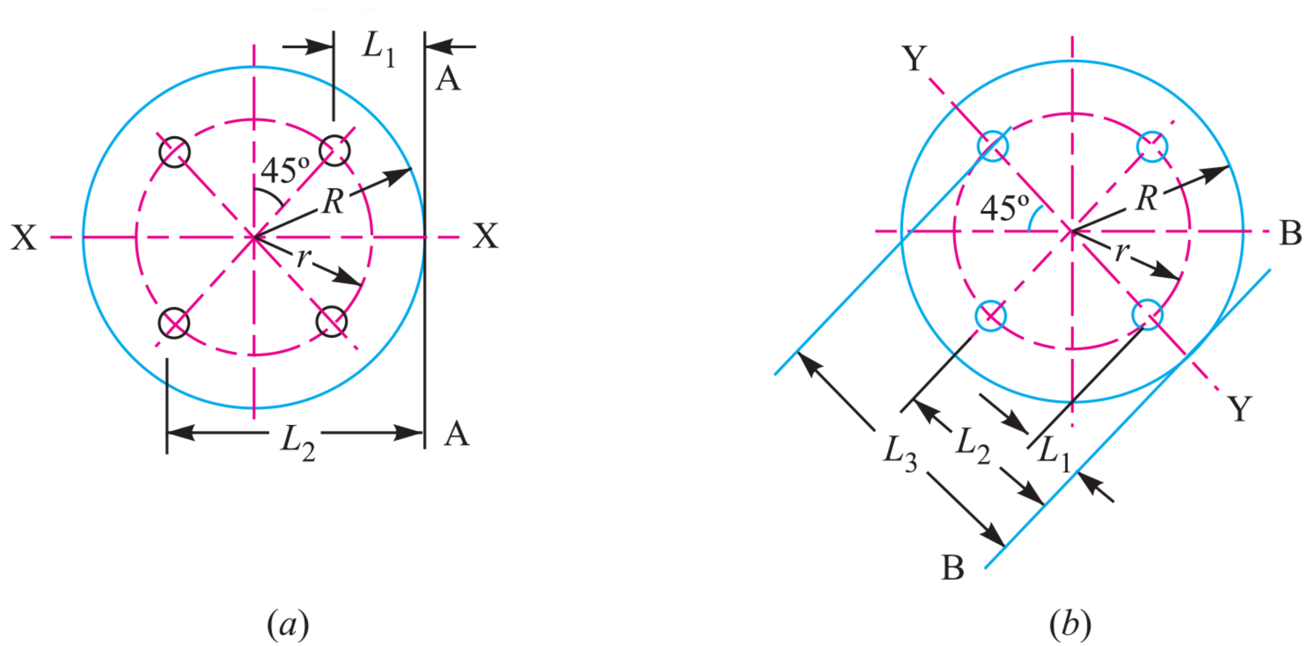


Fig. 11.43

Figure 11.43: 2-Line Overturning Axis Analysis (Line X-X vs Line Y-Y)

SECTION 20: Solid Forged Bracket under Combined Bending, Torsion & Coplanar Shear

System Parameters

Vertical Load: $W = 13.5 \text{ kN} = 13500 \text{ N}$

Eccentricity: $e = 250 \text{ mm}$

Stresses: Tensile $\sigma_t = 110 \text{ MPa}$, Shear $\tau = 65 \text{ MPa}$ | Flange: 4 bolts in square layout

Design Protocol

Part 1: Size arm dia D for combined bending (M) and torsion (T)

Part 2: Size arm dia d for pure bending (M)

Part 3: Calculate top bolt tensile load W_t

Part 4: Calculate maximum resultant shear force W_s on bolts

---

1. Bending & Twisting Moments on Arm D ($\boldsymbol{M, T}$)

Bending moment and torque resolution for diameter D

$$M = W(300 - 25) = 13500 \times 275 = 3.7125 \times 10^6 \text{ N} \cdot \text{mm}$$

$$T = W \cdot e = 13500 \times 250 = 3.375 \times 10^6 \text{ N} \cdot \text{mm}$$

2. Equivalent Twisting Moment (T_e) & Diameter D

Torsional shear stress formula for allowable shear stress $\tau = 65$ MPa

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(3.7125 \times 10^6)^2 + (3.375 \times 10^6)$$

$$5.017 \times 10^6 = \frac{\pi}{16} D^3 \times 65 = 12.76 D^3$$

$$D^3 = \frac{5.017 \times 10^6}{12.76} = 393 \times 10^3$$

$$D = 73.24 \text{ mm}$$

---

3. Bending Moment & Section Modulus for Arm d (M, Z)

Pure bending arm evaluation at depth section

$$M = W\left(250 - \frac{D}{2}\right) = 13500 \times (250 - 37.5) = 2.8688 \times 10^6$$

$$Z = \frac{\pi}{32} d^3 = 0.0982 d^3$$

---

4. Arm Diameter d (Pure Bending)

Flexural tensile stress formula for allowable stress $\sigma_t = 110$ MPa

$$\sigma_t = \frac{M}{Z}$$

$$110 = \frac{2.8688 \times 10^6}{0.0982 d^3} = \frac{29.2 \times 10^6}{d^3}$$

$$d^3 = \frac{29.2 \times 10^6}{110} = 265.5 \times 10^3$$

$$d = 64.3 \text{ mm} \qquad \Rightarrow \textbf{Adopt Diameter } d = 65$$

5. Tilting Edge Distances & Unit Load (L_1, L_2, w)

Overturning moment equilibrium about tilting edge E-E

$$L_1 = 37.5\ \mathrm{mm}, \quad L_2 = 237.5\ \mathrm{mm}$$

$$M_{\mathrm{res}} = 2w(L_1^2 + L_2^2) = 2w((37.5)^2 + (237.5)^2) = 115625w\ \mathrm{N} \cdot$$

$$M_{\mathrm{overturn}} = W \cdot L = 13500 \times 300 = 4.05 \times 10^6\ \mathrm{N \cdot mm}$$

$$w = \frac{4.05 \times 10^6}{115625} = 35.03\ \mathrm{N/mm}$$

6. Tensile Load on Each Top Bolt (W_t)

Secondary tensile force on critical upper bolts

$$W_t = w \cdot L_2$$

$$= 35.03 \times 237.5$$

$$= \mathbf{8320\ N = 8.32\ kN} \quad \mathbf{(Ans.)}$$

7. Primary & Secondary Shear Loads (W_{s1}, W_{s2})

Direct vertical shear and secondary torsional shear per bolt

$$W_{\mathrm{s1}} = \frac{W}{n} = \frac{13500}{4} = 3375\ \mathrm{N}$$

$$r_i = \sqrt{100^2 + 100^2} = 141.4\ \mathrm{mm}$$

$$W_{\mathrm{s2}} = \frac{W \cdot e \cdot r_i}{4r_i^2} = \frac{13500 \times 250 \times 141.4}{4(141.4)^2} = 5967\ \mathrm{N}$$

8. Resultant Shear Force on Bolts 1 & 4 ($\theta = 135^{\text{deg}}$)

Vector combination of primary and secondary shear

$$W_{s(1,4)} = \sqrt{W_{s1}^2 + W_{s2}^2 + 2W_{s1}W_{s2}\cos 135^{\text{deg}}}$$
$$= \sqrt{(3375)^2 + (5967)^2 - 2(3375)(5967)(0.7071)}$$
$$= \mathbf{4303\ N} \quad (\mathbf{Ans.})$$

9. Maximum Shearing Force on Bolts 2 & 3 ($\theta = 45^{\text{deg}}$)

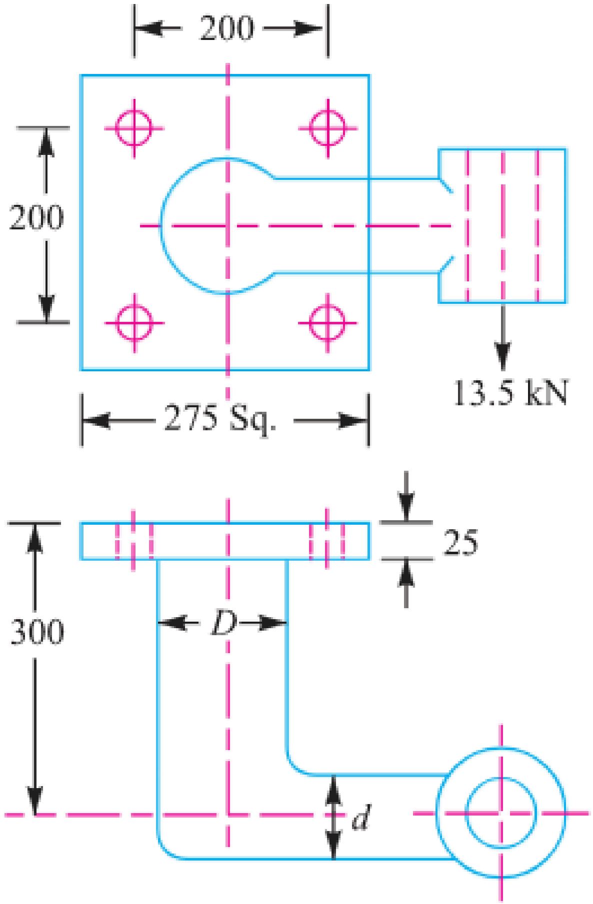
Critical maximum shear force on fasteners

$$W_{s(2,3)} = \sqrt{W_{s1}^2 + W_{s2}^2 + 2W_{s1}W_{s2}\cos 45^{\text{deg}}}$$
$$= \sqrt{(3375)^2 + (5967)^2 + 2(3375)(5967)(0.7071)}$$
$$= \mathbf{8687\ N} \quad (\mathbf{Ans.})$$

10. Design Output

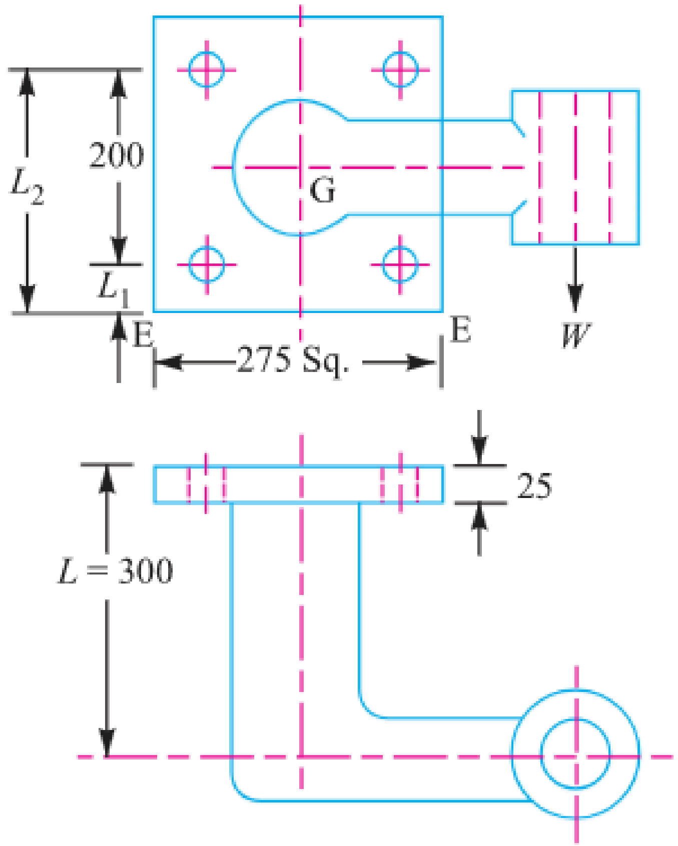
Final Answers: (1) $D = 75$ mm, (2) $d = 65$ mm, (3) Top Bolt $W_t = 8320$ N, (4) Max Shear Force $W_s = 8687$ N

SECTION 20: Solid Forged Bracket Structure — MECHANICAL DIAGRAM



All dimensions in mm.

Fig. 11.45



All dimensions in mm.

Fig. 11.46

Figure 11.45 & 11.46: Solid Forged Bracket Structure & Square Flange Geometry

SECTION 20: Solid Forged Bracket Shear Vector Analysis — MECHANICAL DIAGRAM

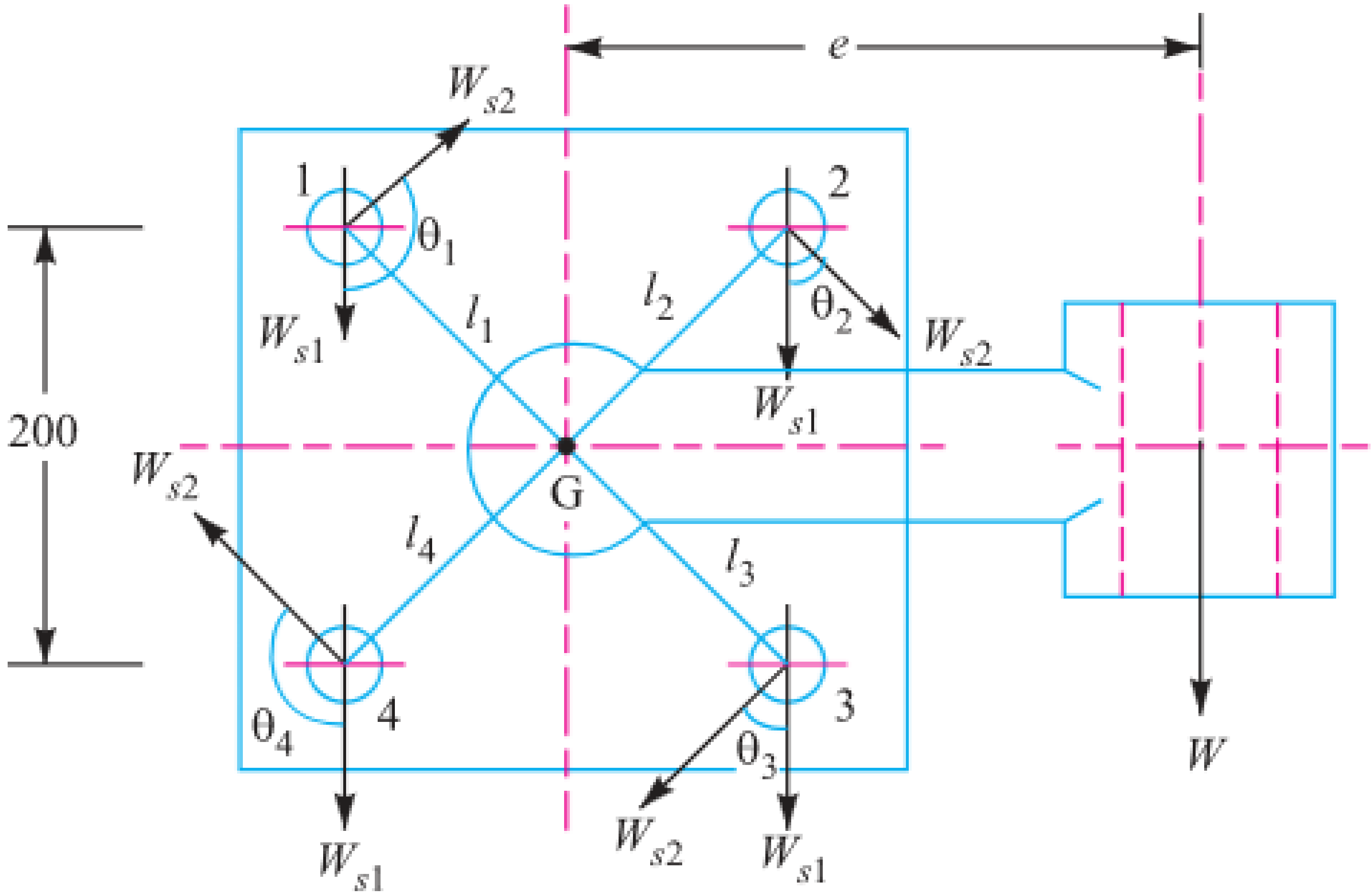


Fig. 11.47

Figure 11.47: Vector Primary & Secondary Shear Force Diagram